Training–Rollout Mismatch in Low-Precision LLM Reinforcement Learning: A Survey

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Abstract: Low-precision rollouts can reduce the cost of LLM reinforcement learning while changing the distribution of experience and the updates learned from it. This survey examines training–rollout mismatch through a common formulation of learner, behavior, proximal, and reference policies. We synthesize twenty-nine studies, distinguishing direct interventions from supporting and boundary evidence. Four classes organize the interventions: computation alignment, numerical fidelity and update visibility, gradient-estimation correction, and policy-update control, with scheduling as a complementary topic. Worked examples connect quantization boundaries, conditional score drift, and signed clipping to learning dynamics. We distinguish controlled numerical probes from online training diagnostics and synthesize how studies measure probability gaps, divergences, weight concentration, and update distortion. Policy versions, decoding, aggregation, masks, and numerical evaluation explain why similarly named curves can describe different objects. We relate studied component combinations to their evidence, implementation dependencies, and total cost, identifying residual errors and open questions in their joint evaluation. Policy quality gained by the complete learning loop provides the common objective for interpreting numerical fidelity, stability, and execution speed.

Keywords: quantized rollout; low-precision reinforcement learning; training–inference mismatch; importance sampling; quantization-aware training.

Literature snapshot: October 20, 2026.

Survey resources: https://github.com/ZunhaiSu/awesome-low-precision-rl

Living survey, under active development. Suggestions, corrections, and research collaborations are welcome via GitHub Issues.

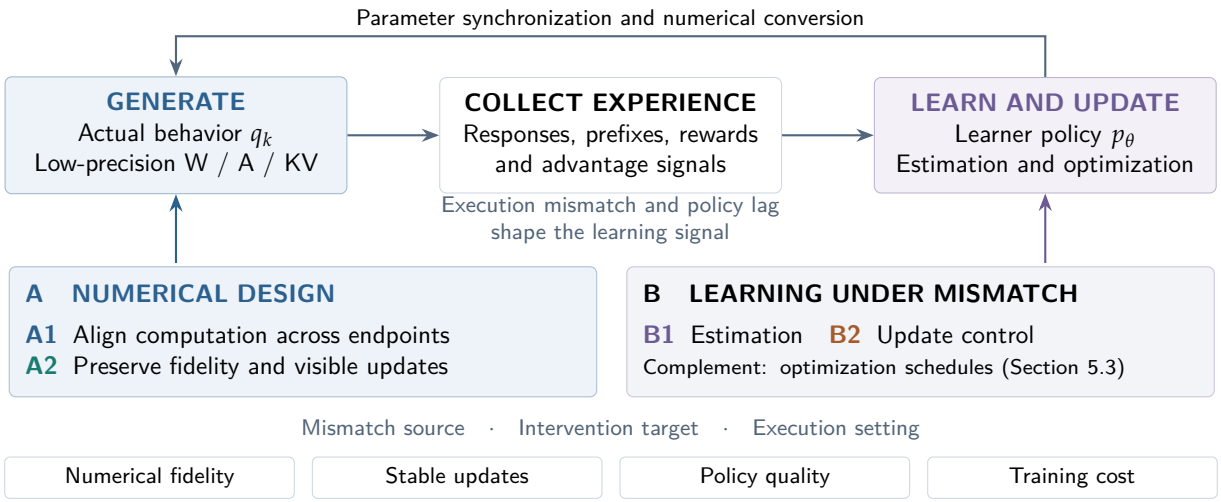


Figure 1: Survey at a glance. Numerical methods act on learner or sampler execution; learning methods change how samples drive updates. Their effects return through the rollout–learning loop. W, A, and KV denote weights, activations, and key–value cache.

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Reading pathways

Foundations	Methods	Evidence
Policy roles, execution mismatch, and its effects on learning.	Numerical interfaces, estimator correction, and update control.	Within-study findings, system costs, and open problems.
Sections 2–3	Sections 4–5	Sections 6–7

Appendix A: surveyed papers · Appendices B–C: sources and assumptions · Appendix D: terminology

Appendix E: mismatch measurements across core studies

1. Introduction and Scope

1.1. Why Low-Precision Rollouts Complicate Policy Optimization

In language-model reinforcement learning, the rollout engine generates the responses and histories used to train the next policy. Lowering its precision can reduce generation cost, but it also changes the distribution of training experience. Even when learner and sampler start from the same parameters, different rounding, kernels, or cached states can produce different token probabilities. This training–rollout mismatch is a numerical instance of training–inference mismatch (TIM).

Its consequences extend beyond the accuracy of a forward pass. A changed probability can alter which response is sampled, how that response is weighted, and whether its gradient survives a clipping rule. The resulting update changes later parameters and rollouts, creating feedback between numerical execution and optimization. Faster decoding is useful when this complete process learns a capable policy at lower cost. Stable curves alone are insufficient: suppressing most updates can prevent divergence while also preventing progress.

Three questions organize the survey. Which computation defines the policy that actually generates experience? Through which mechanism does its discrepancy from the learner affect an update? Which intervention addresses that mechanism, and what benefit has been demonstrated? FlashRL's behavior-probability correction and Jet-RL's coordinated precision flow illustrate two different answers: changing how samples are used and changing the computations that produce their probabilities [1, 3].

Figure 2 organizes methods by the component they change. Numerical methods align computations or preserve useful information; learning methods correct estimation or control updates. A system may combine several such components. Appendix A maps the twenty-nine focal works to their mechanisms and discussion locations.

1.2. Scope and Related Work

This survey focuses on numerical and execution mismatch in LLM reinforcement learning, especially quantized rollout and related low-precision training configurations. It covers a higher-precision learner paired with a lower-precision sampler, as well as low precision on both sides. Weight, activation, and key–value (KV) cache quantization affect different computational interfaces. Policy staleness enters where parameter changes interact with those interfaces.

The twenty-nine focal contributions comprise nineteen method or system studies, nine supporting formulation, theory, or general mismatch studies, and one exploration boundary case. These roles describe their use in the synthesis: MiniRL, for example, develops the formulation and also evaluates FP8 rollout [14]. The literature snapshot is October 20, 2026. Appendix B describes source selection and version handling.

Broader surveys address low-precision LLM training, rollout strategies, and reinforcement learning for reasoning models [33–35]. Primary studies organize the problem through quantization regimes, sources of mismatch, alignment and correction, or collection and cache design [3, 6, 10, 12, 17]. We draw on these distinctions while organizing methods by their intervention targets and evidence by the conditions under which it was measured. General quantization, pretraining, and other rollout accelerators enter where they explain the connection between numerical sampling and learning.

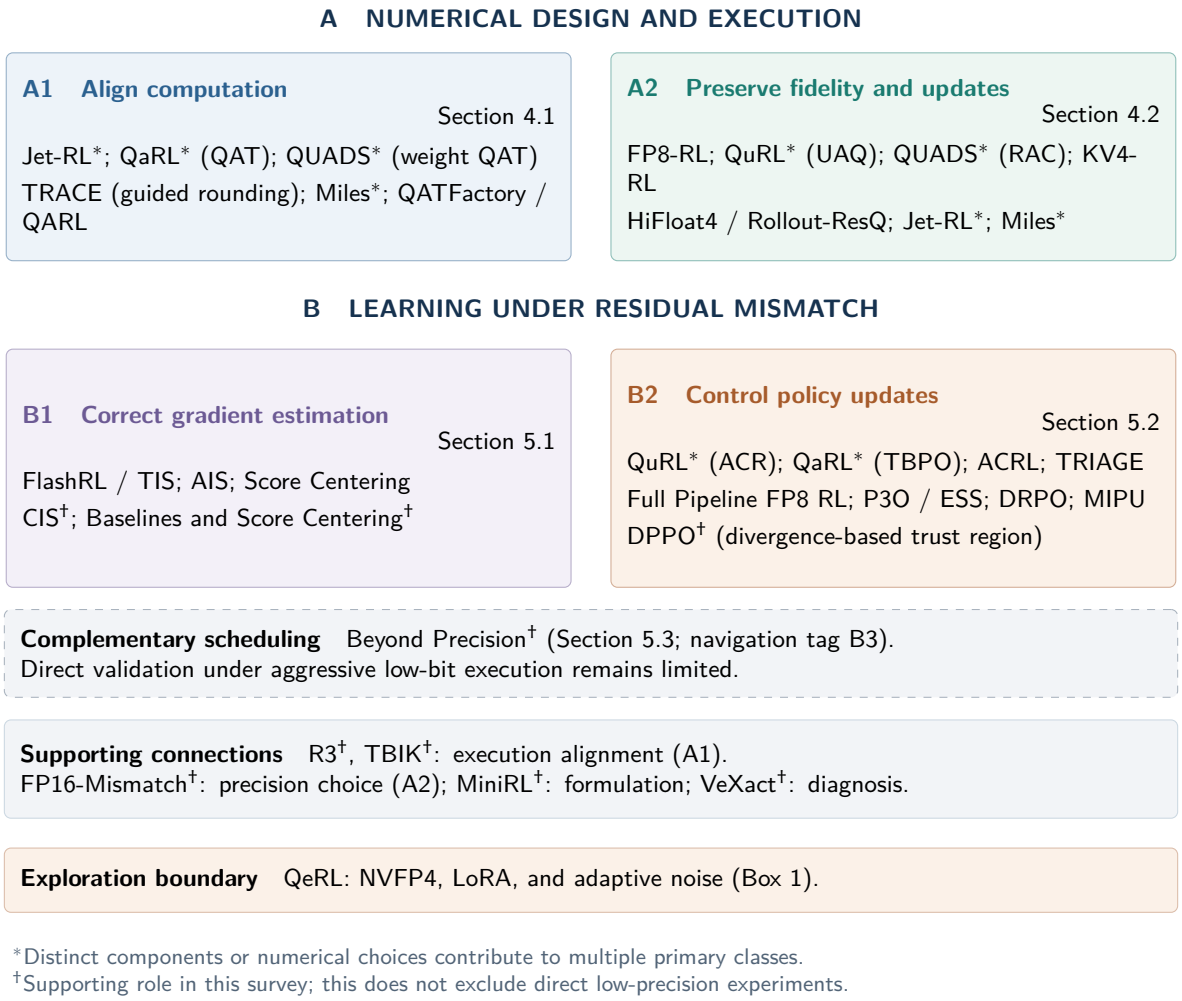


Figure 2: Component-level taxonomy of all twenty-nine focal studies. The four primary classes describe direct interventions; repeated names identify distinct components. Scheduling, supporting connections, and the QeRL exploration boundary are shown separately. Numerical regimes remain independent comparison axes. Appendix A gives the complete section mapping.

1.3. Perspective and Contributions

We connect numerical execution to learning through three contributions: explicit policy roles that separate execution mismatch, parameter staleness, and objective probabilities; a component taxonomy linking failure mechanisms to interventions; and evidence synthesis that distinguishes mismatch measurements and relates them to policy quality and total cost. This perspective accommodates combinations within a system while preserving the conditions of each comparison.

Sections 2–3 establish the background and mechanisms; Sections 4–5 compare interventions. Section 6 synthesizes quality and cost, leading to the research questions in Section 7.

2. Background and Problem Formulation

2.1. Learner, Behavior, Proximal, and Reference Policies

Let x be a prompt, $y_{1:T}$ a response, and $h_t = (x, y_{<t})$ its prefix. A policy is specified by both a parameter state and the computation used to evaluate it. Write $p_\theta(a \mid h)$ for the current learner policy and $p_k(a \mid h)$ for the same learner-side path at sampling snapshot θ_k.

The behavior policy $q_k(a \mid h)$ is the distribution that actually sampled the token. It includes the rollout engine, numerical representation, and sampling transformation. Temperature and vocabulary truncation therefore belong to its definition: a raw-model log-probability returned by an engine need not equal the probability after decoding transformations.

A proximal policy p_{prox} supplies the denominator of a policy-update constraint. A reference policy p_{ref} supplies a regularization target. These are different roles: a frozen pre-update learner may be proximal, while a much earlier policy is the KL reference. Neither is automatically the policy that generated every token. PPO and GRPO provide familiar contexts for these distinctions [30, 31]; mismatch-aware formulations require them to remain explicit [1, 14].

A running GRPO update. Consider a prompt for which the sampler generates G responses. A task-dependent scorer assigns reward R_i to response i; in RL with verifiable rewards, a correctness checker can supply this score without a learned reward model. A common group-relative advantage is $A_i = (R_i - \bar{R})/(\sigma_R + \epsilon_A)$, where $\bar{R}$ and σ_R are the group mean and standard deviation, and $\epsilon_A > 0$ stabilizes the normalization. For binary rewards $(1, 1, 0, 0)$, the advantages are approximately $(1, 1, -1, -1)$. The advantage is held fixed during optimization and is typically shared by the valid tokens of the response. Some variants omit standard-deviation normalization [31].

At the learning stage, a representative clipped objective on this fixed group is

$$\begin{aligned} \widehat{J}(\theta) &= \frac{1}{G}\sum_{i=1}^{G}\frac{1}{T_i}\sum_{t=1}^{T_i}\left[\ell(r_{i,t}, A_i) - \beta \mathcal{K}_{i,t}(\theta)\right], \\ \ell(r, A) &= \min\{rA, \text{clip}(r, 1-\epsilon, 1+\epsilon)A\}. \end{aligned} \tag{1}$$

where T_i counts valid response tokens, $r_{i,t} = p_\theta(y_{i,t} \mid h_{i,t})/p_{\text{prox}}(y_{i,t} \mid h_{i,t})$, ϵ sets the clipping interval, and β controls optional reference regularization. Here $\mathcal{K}_{i,t} = D_{\text{KL}}(p_\theta(\cdot \mid h_{i,t}) \| p_{\text{ref}}(\cdot \mid h_{i,t}))$ denotes the conditional KL; implementations can instead use sampled estimators. Maximizing the surrogate encourages positive-advantage actions and discourages negative-advantage actions. Clipping limits the incentive for excessive changes relative to the proximal policy, while the reference supplies a separate regularization target.

After the update, weights are synchronized and converted for the next rollout. With matched execution and decoding, the pre-update learner can also be the behavior policy: $q = p_{\text{prox}}$. Quantized rollout breaks this identification even before an update. At $p_\theta = p_{\text{prox}}$, the proximal ratio is one, but p_{prox}/q can already differ from one. Section 5.1 shows how an explicit behavior correction can be separated from the update ratio. This loop locates the survey's interventions: numerical design changes the computations, estimator correction changes the use of sampled experience, and update control changes the accepted learning step.

The policy being improved must also be specified. MIPU distinguishes a better learner-side candidate from a better inference policy after conversion and synchronization [23]; the latter is what generates the next experience.

Table 1: Policies and quantities used throughout the survey. Execution paths are part of policy identity.

Symbol	Meaning
p_θ	Current learner policy under its specified execution path.
p_k	Learner-side policy evaluated at sampling snapshot θ_k.
q_k	Actual behavior policy that sampled the token.
p_{prox}	Policy used to define the proximal update ratio.
p_{ref}	Policy used for KL regularization; not the behavior denominator.
Δ_k	Same-snapshot log-probability difference, $\log p_k - \log q_k$.
d_θ	Current learner–behavior gap, $\log p_\theta - \log q_k$.
A_t	Advantage or other policy-gradient learning signal.
$s_\theta(a,h)$	Score vector, $\nabla_\theta \log p_\theta(a \mid h)$.
$c_t, \bar{c}_t$	Behavior-to-proximal correction and its truncated or masked form.
$r_t(\theta)$	Proximal update ratio, $p_\theta / p_{\text{prox}}$.
$W_\theta(y \mid x)$	Complete-response likelihood ratio, P_θ / Q.
N, sg	Valid-token count and stop-gradient operator in Equation (10).

2.2. Execution Mismatch versus Policy Staleness

At a fixed prefix and token with positive relevant probabilities,

$$\frac{p_\theta(a \mid h)}{q_k(a \mid h)} = \underbrace{\left(\frac{p_\theta(a \mid h)}{p_k(a \mid h)}\right)}_{\text{parameter change / lag}} \times \underbrace{\left(\frac{p_k(a \mid h)}{q_k(a \mid h)}\right)}_{\substack{\text{same-snapshot} \\ \text{execution / decoding}}} . \tag{2}$$

The first factor captures parameter change since collection, including asynchronous version lag; the factorization requires no independence assumption. The second compares execution at the sampling snapshot. It isolates numerical differences when decoding is matched and also includes decoding differences when temperature or top-p filtering changes. The intermediate policy p_k is a conceptual reference and need not be retained by the implementation. In partial rollout, a response can span collection boundaries, so $k = k(t)$ may vary across tokens. If caches survive synchronization, retained entries encode earlier computations. Their provenance records the versions and numerical paths that produced them; a current checkpoint identifier alone does not specify this behavior state. The notation q_k abbreviates that stateful rollout path.

Figure 3 separates these snapshot comparisons from the two policy roles used by the loss. A proximal anchor and a KL reference can be chosen independently of the actual sampler.

Two limiting cases make the distinction concrete. At the sampling snapshot, the learner-side parameter-change factor is one while execution mismatch can remain. With matched execution and decoding, the second factor can instead be one while policy lag remains. A numerical intervention and a staleness correction therefore solve different parts of the same ratio.

Probability ratios also require support conditions. A token observed from q_k has a defined denominator, but that alone does not establish that q_k covers every action with positive target probability. Truncation sampling can violate this condition. Similarly, a token-level change of measure is not

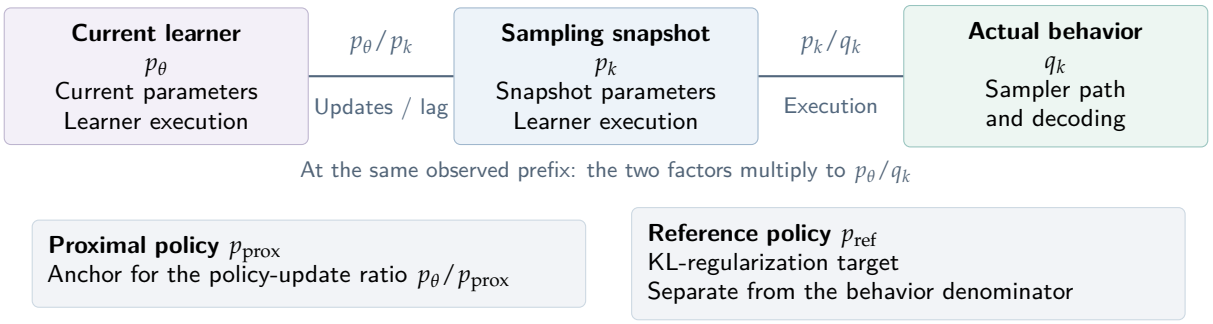


Figure 3: Policy identities and loss roles. Horizontal links denote the ratio comparisons in Equation (2), rather than data flow. Execution differences isolate numerical mismatch when decoding is matched. The lower cards identify separate roles in optimization.

automatically a correction for the prefix distribution. At fixed h,

$$\mathbb{E}_{a\sim q_k(\cdot|h)}\left[\frac{p_\theta(a \mid h)}{q_k(a \mid h)} f(a,h)\right] = \mathbb{E}_{a\sim p_\theta(\cdot|h)}[f(a,h)] \tag{3}$$

holds under support and integrability assumptions. It says nothing by itself about whether the outer distribution over h is the desired one. It also does not undo changes to group-relative advantages induced by collecting different responses.

A token-level correction adjusts the action distribution at the observed prefix; it does not reconstruct histories that the sampler visited with a different frequency. Figure 13 gives a two-step example, and Appendix C states the corresponding full-response identities. The distinction matters for group-relative RL because changing the response distribution also changes the groups from which advantages are computed [10].

2.3. A Component-Level Taxonomy of Interventions

Three complementary views locate a contribution. A *source* identifies why probabilities differ, such as numerical execution, routing, or parameter lag. An *intervention target* identifies what a component changes. An *execution setting* identifies where it was evaluated, including W/A/KV precision, adaptation, and collection. Sources motivate the problem formulation; targets define the classes below; settings condition the evidence. One class can address several sources and appear in several settings.

The taxonomy follows the path from numerical computation to a learning update. **A: numerical design** changes the representation or execution of the policy. A1 asks whether learner and sampler evaluate corresponding computations more consistently; A2 asks whether low precision preserves the information and parameter changes needed for learning. **B: learning under residual mismatch** changes how collected experience produces the next policy. B1 addresses the estimation of a target gradient; B2 controls the influence or acceptance of an update. Table 2 states the decision question for each class.

The distinction is operational rather than a partition of implementation primitives. Both B1 and B2 can multiply a loss by a weight. A B1 component derives that weight or correction from a target expectation; a B2 component uses it to constrain, select, or accept updates. Truncation can make a B1 estimator biased, and a B2 control can also alter estimation. The assignment identifies

the component's direct target, not every effect it has. A loss containing an importance ratio can still contribute a new update constraint: DPPO and DRPO belong to B2 through their trust-region geometry. Score Centering belongs to B1 through its correction of a conditional score expectation [10, 22, 29].

A paper can therefore make coequal contributions in several classes. QaRL combines aligned QAT (A1) with TBPO (B2); QuRL combines update-aware quantization (A2) with adaptive clipping (B2); QUADS combines weight alignment (A1) with activation compensation (A2) [5, 6, 8]. An adopted correction, such as TIS in a numerical system, is distinguished from the system's proposed contribution. Scheduling acts across these choices by changing the scale of learning over time; Section 5.3 treats it as a complementary topic, with B3 retained as a navigation tag.

Table 2: Intervention classes defined by their direct questions. The component map in Figure 2 identifies the associated works.

Class	Decision question	What changes	Key dependency
A1	Can the two execution paths agree more closely?	Operators, representations, or realized numerical decisions.	Correspondence between computations.
A2	Is useful information preserved at low precision?	Scales, residuals, update resolution, or protected states.	Identification of sensitive information.
B1	How should behavior samples estimate a target gradient?	Importance weights or conditional score expectations.	Target distribution and available behavior information.
B2	Which updates should influence the next policy?	Constraints, contribution weights, or candidate acceptance.	Learning signal and control criterion.
B3	When and how strongly should learning proceed?	Complementary optimization schedule.	A signal for adaptation over time.

2.4. Numerical Regimes and Contribution Roles

Quantization maps values to a discrete set using a specified format, scale, and rounding rule. Weight quantization changes parameters that can be shared within a sampling version. Activation quantization acts on values recomputed during each forward pass. KV quantization acts on attention state retained across token positions. These targets differ in when their errors arise and how long those errors remain in the computation.

Post-training quantization applies a representation to a learned model without adapting that model through the target quantizer. Quantization-aware training (QAT) includes quantization effects during learning. A fake-quantized forward quantizes and dequantizes operands to emulate low-bit values, while its matrix multiplications can remain at higher precision. A straight-through estimator can supply a surrogate derivative through rounding, as in QUADS [8]. Native low-precision kernels instead perform the specified arithmetic directly. QAT describes an adaptation procedure, so its name alone specifies neither acceleration nor the precision of backward computation, master weights, or optimizer state.

Static PTQ and QAT do not exhaust the ways a quantized sampler can participate in RL. QuRL repeatedly quantizes the sampler while updating a higher-precision learner from the experience it generates; its parameters adapt indirectly to quantized rollout without an explicit learner-side QAT path [5]. Low-rank versus full-parameter updates and native versus simulated arithmetic are further

independent choices. These distinctions avoid treating QAT, importance correction, and parameter-efficient adaptation as competing method labels.

Architecture and collection protocol supply additional setting axes. The catalog records contribution roles and tested regimes independently: integrating a correction into a system is distinct from introducing it.

3. How Execution Mismatch Affects Learning

3.1. Sources and Propagation of Execution Discrepancy

Scale selection, rounding, accumulation, kernel choice, batch shape, and cached state can all alter a numerical policy's output distribution. Two paths using the same nominal format can consequently disagree. FP16-based analysis illustrates the role of precision selection, while R3 exposes the effect of discrete MoE routing decisions [15, 16]. Routing disagreement can also arise independently of quantization.

Cross-TP deterministic inference isolates another execution source: changing the reduction order can change probabilities even when parameters and nominal precision match. TBIK coordinates reduction trees across tensor-parallel sizes, providing a matched-execution reference rather than a low-bit accelerator [28]. At more aggressive precision, HiFloat4 studies outlier-driven underflow in rollout activations; its precision ablations connect this local information loss to degraded RL outcomes [24].

Shared-prefix comparison holds the token history fixed and isolates differences between conditional distributions. Free-running comparison allows different samples to change the histories visited later. These settings expose different stages of propagation: a small conditional error can redirect generation, while a favorable shared-prefix average can conceal disagreement on histories reached by only one policy.

Discretization introduces a further asymmetry. An update can be visible in the learner's parameter space but fail to change the sampler's quantized representation. Conversely, a small change near a rounding boundary can abruptly change a codeword. QuRL studies the former through update-aware quantization; TRACE targets inconsistency in rounding outcomes [5, 13]. The relevant object is therefore not only error magnitude, but its relationship to the evolving parameters and the discretization boundaries.

Forward mismatch is only one part of low-precision learning. A backward kernel or surrogate derivative can change the computed update even when two forward policies closely agree. Jet-RL specifies backward arithmetic, and Miles exposes a dequantized backward option precisely because gradient fidelity is a separate numerical choice [3, 25]. Policy-gap diagnostics should therefore be interpreted alongside the learner's derivative path, rather than used to attribute every low-precision failure to TIM.

Figure 4 organizes the resulting learning effects into relative influence, expected direction, and update selection. These routes can coexist, which explains why one aggregate disagreement statistic can miss the active mechanism.

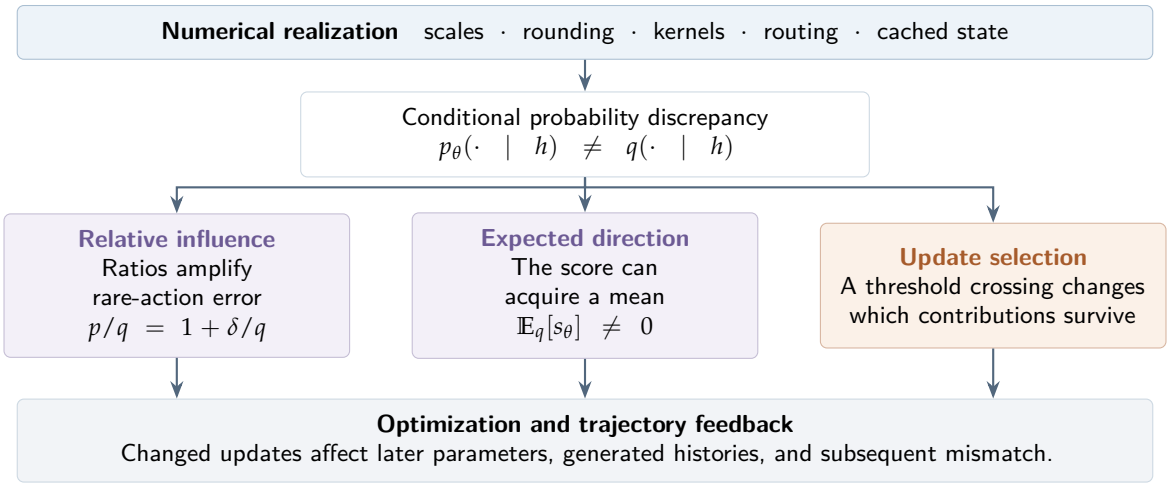


Figure 4: Three routes from execution discrepancy to learning dynamics: relative influence, expected score direction, and update selection. These effects can coexist and feed back through later updates and generated histories. Arrows indicate possible propagation under the policy and estimator assumptions in the text.

3.2. Distorted Estimation and Update Decisions

At an observed prefix, numerical disagreement can change an update through its weight, its expected direction, or the clipping branch it activates. Absolute probability error captures only part of this effect. Writing $p = q + e$ gives $p/q = 1 + e/q$: the same absolute perturbation has greater relative influence when q is small. Tails and confidence-conditioned statistics therefore reveal structure that a mean absolute error can hide. CIS uses this confidence dependence in its correction design [20].

DPPO makes a related distinction between relative ratios and moved probability mass [29]. A large ratio on a rare token need not represent a large distributional shift, while a moderate ratio on a common token can move substantial mass. This motivates separating the geometry of an update constraint from the numerical size of its sampled ratio.

Estimation can also acquire a systematic drift. For a normalized differentiable learner distribution, $\mathbb{E}_{a\sim p_\theta}[s_\theta(a,h)] = 0$. Under behavior sampling, the corresponding expectation need not vanish. At fixed h,

$$\mathbb{E}_q[As_\theta] = \mathrm{Cov}_q(A, s_\theta) + \mathbb{E}_q[A]\mathbb{E}_q[s_\theta]. \tag{4}$$

The expectations are conditional on the prefix and include the sampled action and any random continuation used to form A. Centering advantages across a prompt group need not center them at each prefix. Score Centering uses this distinction to motivate an additive correction [10]: it removes a first-moment drift term while leaving the behavior distribution itself unchanged.

Mismatch additionally affects decisions made by a surrogate objective. In a clipped policy-gradient objective, changing a ratio can change which branch is active and therefore whether a token contributes gradient. Thus, a perturbation can matter through a threshold crossing even when its absolute magnitude appears modest. The full-pipeline FP8 study isolates a failure involving excessive clipping of negative-advantage tokens [11]. This is a different intervention target from merely decreasing the variance of a sampled importance weight.

The sign dependence can be seen without assuming a particular quantizer. Let $\ell(r, A) = \min\{rA, \mathrm{clip}(r, 1-\epsilon, 1+\epsilon)A\}$, with $0 < \epsilon < 1$ and fixed A. Away from the two boundaries,

$$\frac{\partial \ell}{\partial r} = \begin{cases} A\,\mathbf{1}\{r < 1+\epsilon\}, & A > 0, \\ A\,\mathbf{1}\{r > 1-\epsilon\}, & A < 0, \\ 0, & A = 0. \end{cases} \tag{5}$$

Thus the active clipping side depends on the learning signal. For an illustrative objective that clips the combined learner–behavior ratio, a same-snapshot ratio of 0.7 with $\epsilon = 0.2$ already suppresses a negative-advantage token, even before a learner update. This is not the decoupled surrogate in Section 5.1; the example explains why naming the clipped ratio is essential to interpreting a failure or a repair.

Figure 5 makes this sign dependence explicit for the standard surrogate. It also shows why clipping the combined learner–behavior ratio can suppress an update that a proximal ratio would still permit.

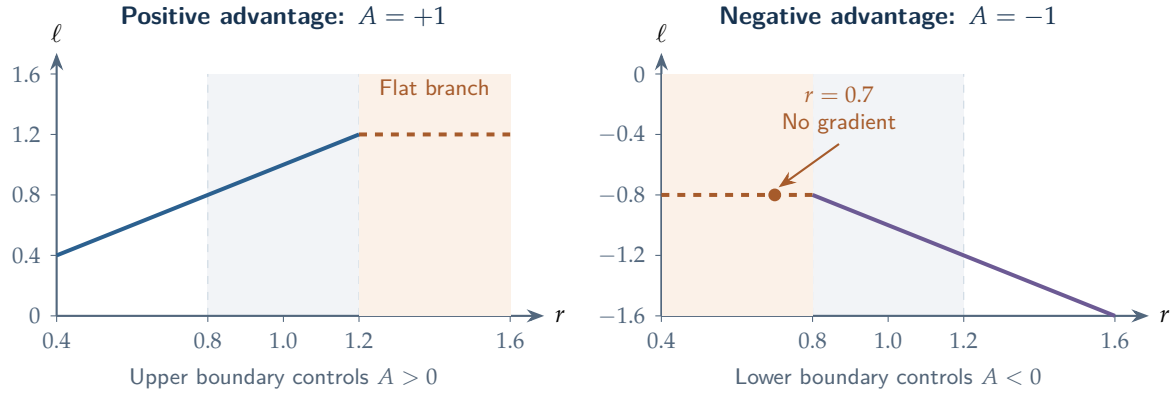


Figure 5: Signed clipping in Equation (5), with $\epsilon = 0.2$ and $A = \pm 1$. The gray interval is $[0.8, 1.2]$; amber segments are flat. At $r = 0.7$, a negative-advantage token contributes no gradient through this objective. The clipped policy ratio determines the interpretation. This is a constructed illustration, not an experimental result.

3.3. Feedback across Optimization Steps

The learning loop turns a local discrepancy into a dynamic process. Updates based on sampled responses change the learner; synchronization transfers those changes to the sampler; subsequent rollouts come from the updated numerical realization. Stability depends on this feedback, not solely on whether one frozen batch has a small error.

The magnitude of a learner–behavior gap alone does not determine how an update changes it. In a simplified local picture, reducing a learner probability that is already below the behavior probability can enlarge their log gap. ACRL and TRIAGE use update-direction information in their controls [9, 12]. For a shared-parameter network, however, one token's update changes other probabilities too. The sign argument supplies local intuition; interactions through shared parameters determine the scope of any stability result.

Optimization scale adds another feedback channel. Beyond Precision studies the relationship between mismatch and optimization dynamics and proposes response-length-triggered learning-rate decay [18]. This connects numerical observations to the scale of optimization, while leaving transfer of the schedule to aggressive FP4 execution open.

3.4. Diagnosing and Measuring Mismatch

Studies share a concern about learner–sampler disagreement, but do not share a measurement protocol. A curve called *mismatch* can describe a probability gap, a KL estimate, concentrated importance weights, or the consequences of a distorted update. Their meaning depends on which policies are compared, what is held fixed, and how observations are aggregated. Table 3 separates their roles; Appendix E maps the reported measurements across all nineteen core studies.

Specify the comparison before the distance. Two measurement regimes answer complementary questions. A *controlled numerical probe* fixes the parameter snapshot and token history, then compares specified execution paths. An *online learning diagnostic* compares the current learner with recorded behavior as training evolves. The latter can include parameter change through Equation (2); independently generated continuations additionally visit different histories. Figure 6 separates these regimes. Raw-model probabilities, actual post-decoding probabilities, and scores recomputed after generation remain distinct comparison objects. VeXact illustrates how a matched-execution reference helps attribute changes in gradients and learning [19].

KV4-RL provides a particularly explicit conditional probe: it computes full-vocabulary $D_{\mathrm{KL}}(p_{\mathrm{BF16}} \| q_{\mathrm{quant}})$ at identical BF16-generated prefixes from the same checkpoint, before temperature or truncation [17]. It therefore measures conditional KV distortion, while leaving trajectory selection outside the metric. TRIAGE shows why conditioning also matters within one study: its 30B characterization replays sampled MoE routes, whereas its 4B run measures the ordinary learner–sampler gap. The paper explicitly avoids comparing their raw magnitudes [12].

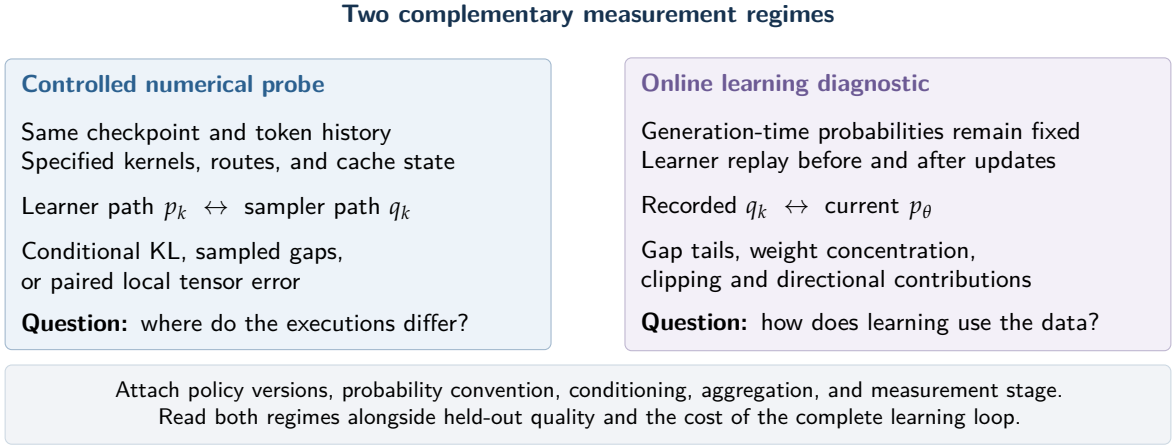


Figure 6: A measurement distinction synthesized from shared-prefix numerical probes and training-time diagnostics [12, 17, 20]. The two sides define different estimands. A controlled probe isolates only the execution factors it actually holds fixed; an online gap can include policy lag.

Probability gaps and KL answer different questions. For a specified pair of normalized conditional policies p, q, let $d_t = \log p(a_t \mid h_t) - \log q(a_t \mid h_t)$ on sampled actions $a_t \sim q(\cdot \mid h_t)$. Common summaries include

$$G_{\text{abs}} = \frac{1}{N}\sum_t |d_t|, \qquad \widehat{K}_1 = -\frac{1}{N}\sum_t d_t, \qquad \widehat{K}_3 = \frac{1}{N}\sum_t (e^{d_t} - d_t - 1). \tag{6}$$

Here N counts the measured valid tokens. G_{abs} measures disagreement on observed actions and removes its sign. For common support and finite expectations, both KL estimators target an average conditional $D_{\text{KL}}(q\|p)$ over the observed prefixes. Their finite-sample behavior differs: $\widehat{K}_1$ can be negative, while each $\widehat{K}_3$ summand is nonnegative. Neither formula is the reverse-direction $D_{\text{KL}}(p\|q)$ or a full-vocabulary calculation at every prefix. A KL to a fixed reference policy serves a separate regularization role.

The distinctions appear in practice. TRIAGE and AIS report mean absolute log gaps; TRACE reports log-gap maxima and minima; QUADS tracks a maximum gap; QuRL reports behavior-to-proximal KL and a maximum ratio; MIPU explicitly uses sampled-token K3 [5, 7, 8, 12, 13, 23]. Extremes and tails reveal effects that a mean can hide, but depend on the size and composition of the measured population. Matching a metric name is only the beginning of a valid comparison.

Table 3: Complementary levels of mismatch diagnosis. Each level answers a different question; outcomes require evidence connecting them to the preceding mechanisms.

Question	Representative measurements	Interpretation boundary
Where do computations disagree?	Shared-prefix KL; sampled-action gaps and tails; paired activation error [8, 12, 13, 17].	Specify checkpoint, prefix, routing, probability convention, and tensor location.
How reliable is the estimator?	Raw or capped weight concentration; ESS; conditional score mean; variance amplification [7, 10, 21].	Concentration and drift are different effects. A theoretical estimand is not automatically a logged statistic.
Which updates change?	Sign-separated clipping; directional mass and segment localization; candidate acceptance [11, 12, 23].	Requires the objective, fixed advantages, and the relevant update or synchronization stage.
Does learning remain useful?	Reward, held-out quality, entropy, failure time, and total cost.	Training outcomes assess consequences; they do not uniquely identify numerical mismatch.

Measure the route from disagreement to learning. Normalized empirical effective sample size (ESS) summarizes weight concentration:

$$\text{ESS}_{\text{norm}}(w) = \frac{(\sum_t w_t)^2}{N\sum_t w_t^2}. \tag{7}$$

P3O uses current-to-behavior ratios, whereas AIS measures the reliability of truncated weights [7, 21]. Their ESS values need not agree on the same batch. Uniform weights give an empirical ESS of one even if their common value differs from one; high ESS alone therefore does not establish calibrated ratios or a faithful gradient. Score Centering addresses another object, the conditional score expectation, whose exact correction and practical reconstruction are distinguished in Section 5.1 [10].

Update diagnostics go further by asking which useful contributions survive. The full-pipeline FP8 study uses BF16 shadow passes to identify tokens whose active clipping branch changes under FP8, separated by advantage sign [11]. This counterfactual *over-clipping* differs from an ordinary clipping fraction. TRIAGE relates signed gaps to directional masses and local concentration, then audits the indicator against realized before/after updates [12]. These measurements connect discrepancy to learning decisions without assuming that every increase in disagreement is harmful.

Comparability requires a measurement specification. A comparable curve needs five pieces of context: the policies and parameter versions; the raw or post-decoding probability convention; the prefixes and execution state held fixed; the vocabulary coverage and aggregation rule; and the measurement stage, population, and uncertainty unit. Zero-advantage masks, route replay, and raw versus truncated weights can materially change the quantity. Appendix E preserves these distinctions instead of filling unreported definitions with a common formula.

Aggregation and measurement can change the conclusion. The mean over all valid tokens weights a response in proportion to its length; averaging within responses first gives each response equal weight. For a constructed example with two responses of lengths 2 and 8 and constant absolute gaps 1 and 0.1, these reductions give 0.28 and 0.55, respectively. Neither is an implementation detail: they describe different populations. In distributed logging, token sums and valid counts must be combined before division to recover a global token mean; P3O explicitly all-reduces the corresponding moments for ESS [21]. A mean of worker means or local maxima generally defines another statistic.

Gating adds a further selection effect. A decreasing mean over retained tokens can reflect removal of the difficult subset, even if disagreement on the original rollout batch has not improved. Report the full valid-token population alongside the intervention-eligible subset, its occupancy, and a defined coefficient- or advantage-weighted mass. TRIAGE's separation of marginal gap statistics, gradient-bearing occupancy, and advantage-weighted masses illustrates why the mask belongs to the definition [12].

Numerical evaluation also needs its own controls. Stored log-probability precision can impose a measurement floor. Near-unit probabilities make log-odds and binary divergences sensitive to saturation and clamping; CIS documents a masking comparison that changes substantially after these operations are evaluated with suitable precision [20]. Logit differences additionally require a convention for their additive offset, because shifting every logit at a prefix by the same constant leaves the policy unchanged. Appendix E collects the resulting reporting guidance and source examples.

The strongest interpretation follows a chain of evidence: a specified numerical disagreement changes an estimator or update decision, which affects learned quality at a stated cost. Smaller gaps establish closer agreement on the measured object. Their value for RL depends on whether useful learning survives. This distinction makes heterogeneous results informative without placing incompatible curves on a shared numerical scale.

Box 1. Quantization and exploration

QeRL combines low-precision execution, low-rank adaptation, and adaptive quantization noise, making exploration an explicit part of the training design [2]. It occupies a boundary of this survey: the numerical perturbation changes the search process as well as agreement between computations. Its results establish a role for quantization noise in that setting, with task utility depending on the induced sampling distribution and the learner's adaptation capacity. They do not establish a benefit from arbitrary cross-engine error.

These mechanisms locate the survey's interventions: the computations that produce disagreement, the estimator that uses sampled experience, and the decisions that determine the next policy. Sections 4 and 5 develop interventions at these points in the learning loop.

4. Numerical Design and Execution Alignment

Numerical design can act on the learner, the sampler, or the interface between them. Alignment brings their computations closer at a shared interface; fidelity control preserves information that a low-precision representation would otherwise lose. The distinction matters because two approximate paths can agree without either becoming more accurate relative to a higher-precision model.

4.1. Aligning Learner and Rollout Computation

Jet-RL treats precision as a property of the computation graph shared by training and rollout [3]. It coordinates corresponding forward operators and quantization granularity instead of placing an FP8 sampler beside an otherwise unchanged learner. Its linear-layer forward and backward GEMMs use FP8 inputs, while gradients transported between operators remain BF16. The relevant alignment is thus more specific than a common format label and more substantial than learner-side fake quantization alone. Its results motivate coordinating the numerical path before asking a correction to absorb all cross-engine differences.

QaRL brings the learner forward closer to the representation used by quantized rollout through rollout-aligned QAT [6]. This numerical component is paired with TBPO, which controls sequence-level updates when low-bit generation produces problematic responses. The two components address different stages of the loop: QAT changes the probabilities computed by the learner, while TBPO changes how sampled responses drive optimization. Their joint use makes QaRL an early example of alignment and update control within one method.

QUADS sharpens the distinction between weights and activations [8]. Weights synchronized across engines can share a quantize–dequantize path. Activations are recomputed from engine-dependent inputs, so applying the same coarse quantizer to both sides can send nearby values to different codewords. QUADS therefore fake-quantizes learner weights while retaining BF16 learner activations, then compensates selected rollout activation errors. Its asymmetric-versus-symmetric ablation shows why matching the list of quantized operands need not improve their realized agreement. The residual activation component is discussed in Section 4.2.

These approaches use three forms of shared information: a coordinated operator configuration, a target representation, or selected numerical values. The distinction leads to a more local alignment question: can the learner use the actual rounding outcome from the rollout that produced a sample?

TRACE aligns realized rounding decisions [13]. A shared FP4 format and scale can still send nearby learner and rollout activations to different codewords. TRACE records rollout-side quantization information and uses it to guide the learner's corresponding quantization site. With exact guidance and fixed rollout scales, it selects from the learner value's two neighboring codewords the one nearest the rollout codeword. Figure 7 illustrates why this can reduce cross-path discrepancy even when it slightly increases the learner's own quantization error.

The practical mechanism approximates this reference with compact code information. By default, TRACE caches one low code bit and the scales from the latter half of the layers, pairing records by trajectory, token position, and quantization site. The reconstruction uses the current learner activation as side information: after normalization by the rollout scale, it selects the nearest E2M1 magnitude whose retained code bits match the record, taking the sign from the learner. The adjacent learner candidates are then compared with this reference. Thus, the record narrows a local choice; it does not independently recover an arbitrary rollout value. Earlier layers can still be quantized without retaining this guidance. The reported adjacency pattern motivates this compression, rather than supplying a universal recovery guarantee.

Precision on the two endpoints remains distinct. The main rollout setting combines NVFP4 expert weights and activations with NVFP4 KV storage in softmax-attention layers; other modules remain BF16. The learner uses rollout-guided QAT, including fake quantization at the corresponding expert and KV sites, with a BF16 backward path. Training-side attention remains in BF16. This is not evidence for native-FP4 trainer acceleration. Routing replay and token-level importance correction address other residual differences, while asynchronous collection can still introduce policy lag. The local rounding property therefore does not imply full-network agreement or removal of staleness [13].

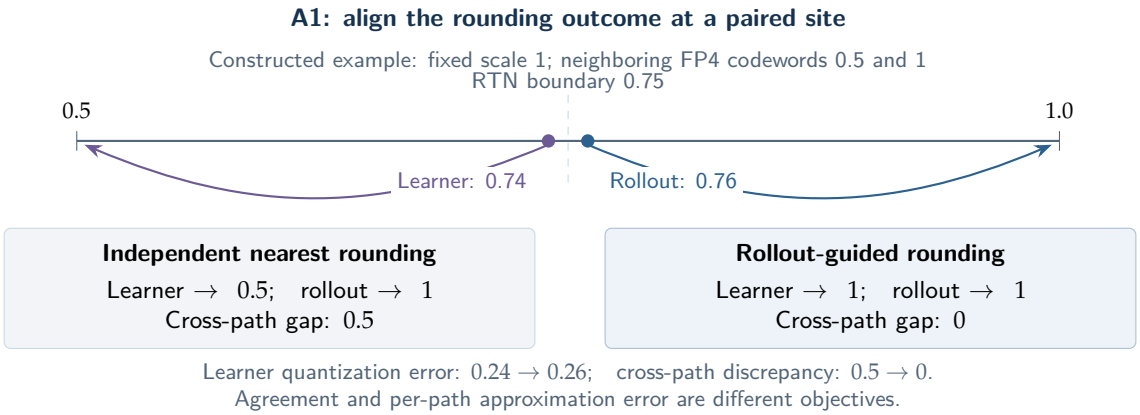


Figure 7: Constructed fixed-scale example of rollout-guided rounding using valid E2M1 codewords. The learner's approximation error increases slightly while its discrepancy from the rollout codeword vanishes. Exact guidance concerns this paired site; TRACE's compact implementation approximates the reference.

System implementations extend alignment across the model lifecycle. Miles coordinates checkpoint conversion, learner forward computation, rollout kernels, and live weight export through shared quantization rules [25]. Its FP8, MXFP8, and NVFP4 recipes retain higher-precision exceptions; MXFP8 and NVFP4 are marked Beta. QATFactory's QARL path pairs native NVFP4 rollout with target-quantization simulation in BF16 training, then exports inference-ready low-bit weights [26]. Here alignment concerns representation and conversion semantics, while training matrix multiplications remain in BF16.

Alignment consequently differs in what must be shared and retained. Jet-RL coordinates a numerical configuration; aligned QAT matches a representation; TRACE additionally transports rollout-specific decisions. Routing replay couples discrete MoE choices through a related principle [15]. TBIK illustrates the stronger requirement of coordinating reduction order across parallel configurations [28]. More specific coupling introduces dependencies on token positions, modules, and parameter versions. Agreement of the complete policies still depends on the computation surrounding the shared interface.

4.2. Controlling Quantization Error and Preserving Parameter Updates

Alignment alone does not ensure that the sampler can express the changes made by training. For an illustrative deterministic quantizer Q, an update can satisfy

$$Q(w + \delta w) = Q(w), \tag{8}$$

even when $\delta w \neq 0$. Visibility depends on the distance to a quantization boundary and on scale behavior, as well as update magnitude. QuRL's Update-Aware Quantization addresses this gap through a one-time invariant rescaling before RL [5]. The real-valued identity $WX = (W/s)(sX)$ preserves a layer's initial output while changing the magnitudes seen by quantization and optimization. The method chooses the rescaling to improve update size relative to quantization noise. Preserving the initial real-valued function does not imply identical later optimization trajectories: the parameterization and gradients have also changed. UAQ therefore targets the sampler's ability to express learning, whereas QuRL's ACR changes which gradients reach the learner through clipping (Section 5.2).

Figure 8 shows a smaller update crossing a boundary while a larger one stays within its cell. Changing the scale moves these boundaries.

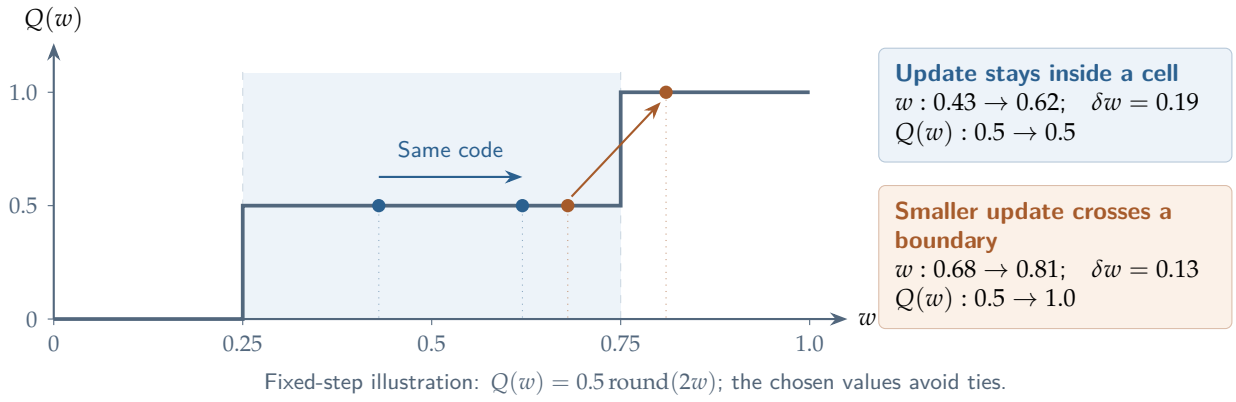


Figure 8: Quantized-update visibility in Equation (8). Under the fixed-step scalar quantizer, the blue update stays inside a cell while the smaller amber update crosses a boundary. Variable scales change these boundaries. This is a constructed illustration, not an experimental result.

QUADS addresses the activation error left after learner-side weight alignment. Residual Activation Compensation restores selected high-error channels around the native W4A4 rollout computation [8]. Weight QAT and residual correction consequently change different quantities: the discretized weights shared with the learner and the rollout activation approximation. The component ablation separates the substantial recovery from weight alignment from the additional gain obtained by compensation. This is a concrete instance of A1 and A2 complementarity.

HiFloat4 / Rollout-ResQ targets rollout activation underflow through sparse residual correction [24]. Comparing learner and sampler precisions isolates a sampling-path error that raising learner precision alone does not resolve. The FP4 experiments use simulation, so they characterize learning under the simulated arithmetic; native-FP4 acceleration remains unmeasured.

FP8-RL maintains numerical fidelity as the policy changes through blockwise W8A8 rollout, scale recalibration for FP8 attention and KV execution, and dynamic weight synchronization [4]. It integrates these components with adopted TIS/MIS correction. Its v2 experiments also explore end-to-end FP8, so learner precision varies across the paper's experimental settings.

Miles exposes fidelity choices within an integrated precision recipe: selected tensors remain in BF16, and an optional dequantized backward path trades training throughput for gradient stability without changing the quantized forward values [25]. Such exceptions belong to the numerical configuration; they do not imply that every operation uses the headline format.

KV4-RL allocates fidelity across positions instead of raising precision uniformly [17]. INT4 KV storage is supplemented by bounded INT8 protection for attention-sink and recent positions. Its retained TIS recipe makes the distinction between sampler fidelity and conditional correction especially clear: protecting the cache changes the data-generating computation even when the correction is held fixed. The experiments use colocated, quota-based partial collection, with generation and optimization alternating; responses may span collection boundaries.

These methods preserve different information needed by learning. Parameter rescaling protects update visibility, residual compensation reduces activation error, recalibration follows distributional change, and positional protection targets sensitive cache states. Scaling, rotation, and residual correction are useful categories of quantization technique, as discussed by HiFloat4 [24]. For RL, the additional question is when their parameters or protected regions are chosen and whether they track an evolving policy. The transformation type alone does not determine compatibility with repeated rollout and optimization.

4.3. Residual Mismatch and Limits of Alignment

Alignment and fidelity constrain different comparisons: agreement between endpoints and faithfulness to a higher-precision computation. Shared weights can coexist with different activations, and a common quantizer with different surrounding kernels. Each result is therefore conditional on the shared interface, parameter snapshot, and input history.

Residual differences connect numerical design to learning correction. A more faithful sampler may still lag behind the learner, and conditional agreement at observed prefixes does not restore a response distribution changed by earlier decisions. A learning intervention addresses how these remaining differences affect estimation or the resulting update.

Maintaining a shared numerical interface requires information as well as arithmetic. Operator choices require coordination, QAT depends on matching quantizer semantics, and rounding guidance requires storage and correspondence with learner states [3, 6, 8, 13]. Fidelity control instead uses calibration, compensation, or selective execution to preserve sensitive information [4, 5, 17]. Table 4 compares these dependencies and the discrepancies each leaves unresolved.

Residual disagreement makes the use of collected experience the next design choice: which expectation to estimate and which policy changes to permit.

Table 4: Numerical strategies, the information they share or preserve, and their remaining execution differences.

Strategy	Shared or protected object	Information dependency	Residual discrepancy
Precision flow	Corresponding operator precision and granularity	Coordinated numerical configuration	Engine and surrounding-kernel differences.
Aligned QAT	Learner representation corresponding to rollout	Matching quantizer semantics	Native versus simulated arithmetic; unmatched activations.
Guided rounding	Realized rollout quantization decisions	Stored and matched rollout guidance	Unshared modules and downstream propagation.
Fidelity control	Scales, update resolution, residuals, or protected states	Calibration, compensation, or selective execution	Other execution paths and policy staleness.

5. Learning under Residual Mismatch

Residual mismatch enters learning through both estimation and optimization. Estimator correction changes how behavior samples represent a target expectation; update control regulates their influence; scheduling changes the scale of adaptation over time. These objectives lead to different trade-offs among bias, variance, progress, and stability.

5.1. Correcting Gradient Estimation

Multiplicative correction and truncation. Importance sampling supplies a conditional change of measure under the assumptions of Equation (3). Practical surrogate objectives usually distinguish the mismatch correction from the proximal update ratio. With frozen p_{prox} and recorded behavior q, define

$$c_t = \frac{p_{\text{prox}}(a_t \mid h_t)}{q(a_t \mid h_t)}, \qquad r_t(\theta) = \frac{p_\theta(a_t \mid h_t)}{p_{\text{prox}}(a_t \mid h_t)}. \tag{9}$$

A representative decoupled surrogate is

$$\widehat{L}(\theta) = \frac{1}{N} \sum_t \text{sg}(\bar{c}_t) \min[r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t], \tag{10}$$

where $\bar{c}_t$ is a possibly truncated correction, the sampled advantages are held fixed, and sg denotes no differentiation through the correction weight. Before truncation and clipping, $c_t r_t = p_\theta/q$: the product compares the current learner with the actual sampler. Keeping the two factors separate makes their purposes explicit. Bounding c_t limits the correction for how a token was collected; clipping r_t limits the incentive to change its probability during the update. Individual objectives differ in how they combine these operations. An upper-truncated correction, for example, takes $\bar{c}_t = \min(c_t, C)$, whereas masking can discard selected ratios. Here N counts valid sampled tokens. Response-level averaging and length normalization assign different relative weights, so the reduction rule is part of the estimator being compared.

FlashRL's truncated importance sampling (TIS) provides a basic bounded correction using actual rollout probabilities [1]. MiniRL exemplifies the mismatch-aware distinction between behavior and

proximal policies underlying this notation [14]. FP8-RL adopts token-level TIS and masked importance sampling (MIS) variants, while KV4-RL retains TIS alongside cache protection [4, 17]. The system studies show how the correction operates alongside other numerical components. Truncation bounds extreme influence while changing the expectation; recording an actual behavior denominator supplies the required probability semantics, but does not remove that truncation bias.

Adapting the amount and location of correction. Adaptive importance sampling (AIS) makes correction strength depend on both necessity and reliability [7]. Its effective multiplier has the form $1 + \alpha(\bar{w} - 1)$: $\alpha = 0$ retains the uncorrected estimator, while $\alpha = 1$ applies the truncated importance weight $\bar{w}$. A mean absolute log gap indicates whether correction is needed; capped-weight ESS and advantage-weight variance amplification indicate whether it is reliable. A large mismatch can therefore coexist with weak correction when the weighted estimator is too concentrated or noisy. AIS adapts an estimator mixture rather than the precision or clipping threshold.

Calibrated importance sampling (CIS) changes where truncation acts. Its log-odds displacement separates confidence from cross-engine perturbation: the same displacement can induce very different ratios at low- and high-probability actions [20]. A confidence-dependent cap follows this representation. AIS and CIS thus adapt different objects—mixture strength and truncation location—using different diagnostics. CIS supplies general TIM evidence from BF16 cross-engine experiments; transfer to an FP4 numerical path remains to be evaluated.

Additive score correction. Score Centering addresses a different object from the sampled-token importance weight [10]. Here the *score* is the parameter-gradient vector $s_\theta(a, h) = \nabla_\theta \log p_\theta(a \mid h)$, not the reward assigned to an answer. Its mean is zero under the learner distribution, but can be nonzero under the actual behavior distribution. In Equation (4), that conditional mean produces an additional term proportional to the conditional mean advantage. Group-centered rewards need not remove it because prompt-level centering does not center the signal at every generated prefix.

Score Centering replaces the score at a prefix with

$$\widetilde{s}_\theta(a, h) = s_\theta(a, h) - \mathbb{E}_{b \sim q(\cdot|h)}[s_\theta(b, h)]. \tag{11}$$

With the exact expectation, its mean under q vanishes and the drift term in Equation (4) is removed [10]. The remaining covariance is still measured under q, so this is not full recovery of an on-policy gradient. The reported implementation uses sampler top-k log-probabilities and models the omitted tail using the learner distribution; its default is $k = 128$ [10]. The omitted mass therefore enters through a modeled tail rather than being discarded. The coefficients used to form the correction are detached, and its fidelity depends on that behavior reconstruction.

Figure 9 gives a one-parameter example. Changing the gradient's conditional mean does not require changing the sampler or forcing the learner probabilities to equal it. This separates a correction to the estimator from direct numerical alignment. When combined with importance weighting, the centered object must be the corresponding weighted score; an arbitrary weighted loss with the original unweighted center need not have the intended cancellation [10].

The covariance analysis of baselines and score centering makes estimator granularity central to their composition [27]. For the sequence-level regularized importance estimators studied there, the group-mean baseline already supplies the centering benefit. At token level, the same baseline

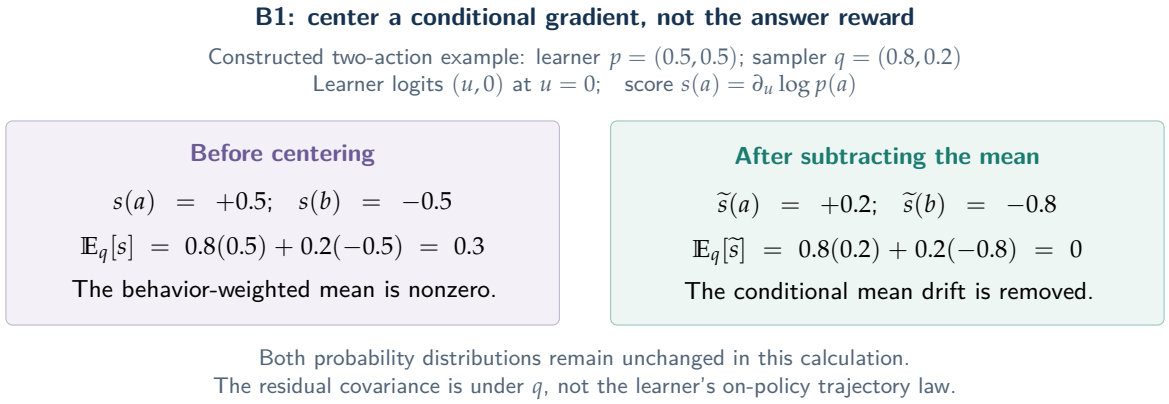


Figure 9: Exact score centering in a constructed scalar-logit example. Sampler probabilities weight the derivatives in Equation (11). Centering changes the conditional gradient mean while leaving both distributions fixed; practical top-k reconstruction adds an approximation.

generally leaves bias under regularized ratios. The useful centering construction therefore depends on the estimand, weighting, and reduction rule.

The methods therefore differ in both the error they target and the information they require. TIS-like rules can use the probability of the sampled action; AIS and CIS adapt the use of such correction. Score Centering additionally reconstructs a distributional expectation over actions at the prefix. A cap approximates a change of measure, whereas a reconstructed center approximates a first moment. The distinction explains why access to sampled-token log probabilities alone is enough for one operation but not for the other.

5.2. Controlling the Influence of Policy Updates

Update control spans five decisions: where to place a boundary, how to adapt it to batch reliability, which divergence geometry to use, which signed contributions to retain, and whether to accept a synchronized candidate. Table 5 separates these decisions from their aggregation units.

Table 5: Control decisions within B2. The unit is the object controlled or the level at which its signal is measured; it does not define a separate primary class.

Decision	Signal or constraint	Unit	Representative components
Boundary placement	Mismatch-aware, trust-band, or reference-calibrated clipping.	Token or response.	QuRL ACR; QaRL TBPO; Calibrated Clipping.
Batch adaptation	ESS determines cap and regularization.	Batch statistics, applied through the objective.	P3O.
Constraint geometry	Divergence-based boundary or smooth regularizer.	Token-level probability shifts.	DPPO; DRPO.
Directional influence	Advantage and discrepancy determine retained or corrective contributions.	Tokens and segments within responses.	ACRL; TRIAGE.
Candidate acceptance	Assessment after inference-side synchronization.	Whole candidate policy.	MIPU.

Restoring or reshaping update constraints. QuRL's Adaptive Clipping Range (ACR) responds to the relation between proximal and behavior probabilities [5]. Its concern is that a fixed clipping rule can interact poorly with the mismatch already present at collection. ACR changes the permitted update region; UAQ, discussed in Section 4.2, instead changes whether the quantized sampler can express the resulting weight update. These two mechanisms distinguish failure to learn an update from failure to transmit it through quantization.

QaRL's Trust-Band Policy Optimization (TBPO) combines negative-sample treatment with a sequence-level trust band [6]. This changes the unit of control: a whole response, rather than an isolated token ratio, determines how an update is constrained. The full-pipeline FP8 method uses a different reference. It matches the lower clipping quantile on negative-advantage tokens to a BF16 run, then adjusts the upper boundary to rebalance positive and negative contributions [11]. Periodic reference passes amortize this calibration, whose target is the objective boundary rather than the quantizer scale.

These controls make the learning signal that survives clipping central to interpretation. Widening a region admits more updates, but a larger active fraction is valuable only when those contributions improve learning. Advantage sign, response reduction, and the comparison policy determine what a clipping statistic means.

Adapting control to batch reliability. P3O uses normalized effective sample size (ESS) of the batch's policy ratios to adapt a score-weight cap and off-policy regularization [21]. Batch reliability thus determines how strongly the update is constrained. Its FP8-rollout experiment studies numerical mismatch directly, while the broader comparisons show dependence on the model and collection setting. AIS adapts the mixture of gradient estimators; P3O uses batch reliability to set update constraints. Similar adaptive machinery consequently serves a different target.

Choosing the constraint geometry. DPPO replaces sampled-ratio clipping with a divergence-based trust region [29]. DRPO retains the Binary-TV geometry but replaces hard masking with a smooth regularizer that supplies corrective gradients outside the boundary [22]. DPPO provides the broader trust-region foundation; DRPO evaluates both FP8-rollout-only and end-to-end FP8 MoE settings. The shared object is the geometry of the permitted update, while the available low-precision evidence differs.

Controlling disagreement through update direction. ACRL uses advantage-guided probability adjustment, with response-level discrepancy relative to a reference controlling token reweighting [9]. Its target is how updates change disagreement, rather than the recovery of a distributional expectation.

TRIAGE connects the advantage sign to the learner–sampler log gap [12]. Using $d_\theta = \log p_\theta - \log q_k$ in this survey's notation, the product $A_i d_{\theta,i,t}$ identifies the sign of a simplified token's own mismatch-energy contribution. Cross-token coupling and optimizer history can change the realized motion, so this is a local directional indicator. Both $A_i > 0, d_\theta > 0$ and $A_i < 0, d_\theta < 0$ are locally amplifying. TRIAGE intervenes selectively on the latter: increasing the probability of a reward-positive response can be legitimate learning even if its gap grows.

The gate first diagnoses 64-position segments of negative-advantage responses from their mean gap and deep negative tail. Within an eligible risky segment, only negative-gap tokens receive an

attenuated raw weight $b_{i,t}$. The weighted response loss is normalized by the sum of its weights. Relative to ordinary response averaging, its effective multiplier is therefore $b_{i,t}/\bar{b}_i$, not simply $b_{i,t}$. Normalization reallocates contributions: for illustrative raw weights $(0.3, 1, 0.3, 1)$, the effective multipliers are approximately $(0.46, 1.54, 0.46, 1.54)$. If every weight is equal, the normalization cancels it. This controls relative influence, not the total gradient norm.

Repair acts on positive-advantage responses with sufficiently negative mean gaps in separate 128-position segments. A one-sided pseudo-Huber penalty adds upward pressure on learner log probabilities. Its derivative bounds the additive corrective coefficient; it does not guarantee a bounded full gradient or contraction for every token. All valid tokens in an active repair segment share that coefficient, including any positive-gap tokens. Gating and repair thus modify different signed contributions rather than uniformly penalizing disagreement [12]. The evaluated recipe preserves native W4A4 forward execution on both endpoints. The documented KV cache is BF16, and the paper reports no KV4 experiment. Applying the same control to KV4 is a transfer hypothesis, not an established result.

Accepting whole updates in the inference policy. MIPU constructs candidates relative to the sampler, then assesses whether a synchronized candidate should become the next inference policy [23]. This separates candidate construction from acceptance after numerical conversion. The practical inference-gap proxy supports acceptance or rollback in the evaluated FP8 runs; a general guarantee of monotonic improvement would require stronger conditions.

The decision and its unit must be read together. Local rules can reveal concentrated risk but rely on noisier statistics; aggregation improves reliability while obscuring location. Whole-update acceptance asks a further question: whether a favorable learner-side candidate remains useful after conversion to the inference policy.

5.3. Adapting the Optimization Schedule

A complementary intervention changes the scale of learning over time. Beyond Precision motivates learning-rate scheduling through the evolution of optimization and mismatch [18]. This is complementary to changing a numerical path or a token weight: it modifies the trajectory of parameter updates themselves.

Schedule adaptation remains less developed for direct low-bit settings than the estimator and update controls above. Its promise depends on preserving progress as well as stability: smaller steps can avoid divergence while delaying useful learning. Comparable quality within a comparable budget is therefore the relevant test of transfer to aggressive quantization.

5.4. Compatibility, Bias, and Failure Modes

Two interventions are complementary when they address residual effects left by one another. Numerical and learning components often meet this description: weight alignment can leave activation error, and a more faithful sampler can still generate off-policy data. The tested combinations in QaRL, QuRL, and QUADS illustrate distinct forms of complementarity [5, 6, 8].

Within the learning objective, compatibility is more demanding. The baseline analysis shows why token-level and sequence-level regularized estimators can require different centering constructions [27]. Weighting, centering, masking, and reduction together define the gradient. Figure 10

Table 6: Studied component combinations and the residual effects they address. A tested bundle does not establish arbitrary interchangeability of its components.

Study	Combined components	Mechanistic connection
QuRL [5]	Update-aware quantization (A2) and adaptive clipping (B2).	Make parameter changes visible to the sampler and reshape the learner's update constraint.
QaRL [6]	Aligned QAT (A1) and TBPO (B2).	Reduce representational disagreement, then constrain problematic response-level updates.
QUADS [8]	Weight QAT (A1) and residual activation compensation (A2).	Weight alignment leaves activation error for a separate numerical component.
KV4-RL [17]	Selective cache protection (A2) with retained TIS.	Improve the data-generating path while holding a bounded conditional correction in the recipe.
TRACE [13]	Guided QAT (A1), routing replay, and token-ratio correction.	Couple rounding outcomes while separately addressing route choices and residual off-policy effects.
Score Centering [10]	Centering composed with TIS or MIS (B1).	Center the corresponding weighted score; reconstruction and regularization determine the remaining bias.

distinguishes the target of an estimator correction from the signed contributions selected by update control.

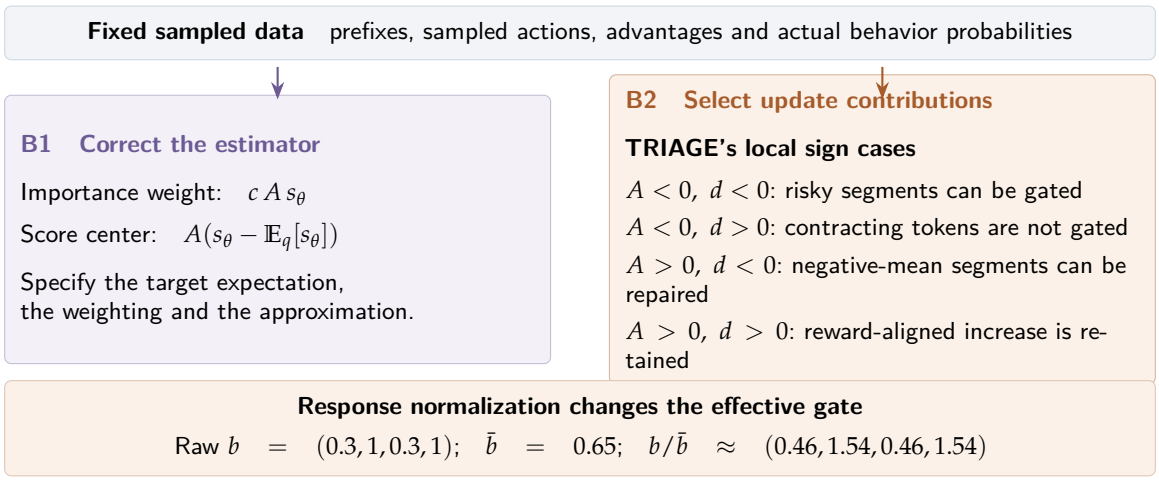


Figure 10: Estimator correction and update control. TRIAGE's sign cases use $d = d_\theta$ together with segment conditions. Illustrative raw weights show how response normalization changes effective influence; they do not represent a diagnosed segment. Shared-parameter interactions limit token-wise directional predictions.

Score Centering and TRIAGE exemplify this distinction. Centering removes a conditional mean term in the score; gating and repair change selected updates according to their advantage and log-gap signs. Neither mechanism is a substitute for the other by definition, and their different source experiments do not establish a preferred order of composition [10, 12].

Composition consequently requires three checks. First, the probability records and execution states must refer to the policies used by every component. Second, weighting, centering, masking, and normalization must define the intended gradient: an importance-weighted score needs its corresponding weighted center [10]. Third, the combined system must retain useful updates at an acceptable cost. Successive masks can suppress the same examples, and guidance or reference passes can consume the generation savings. Table 6 identifies combinations supported by the reviewed studies; adding an untested component remains a separate empirical question.

6. Evidence Synthesis and System Trade-offs

The published evidence supports three recurring conclusions: matching numerical formats is weaker than matching computations; controlling disagreement is useful only when informative learning survives; and faster generation delivers a smaller or different benefit at the level of the complete RL loop. We develop these conclusions through source-specific comparisons, then examine the information and execution costs of the interventions. Models, numerical paths, and evaluation rules remain attached to each result.

6.1. Experimental Regimes

Asymmetric learner–sampler precision exposes differences between numerical representations. Low precision on both sides makes the consistency of their execution paths and the resulting update especially salient. KV quantization acts on retained state, whereas weight and activation quantization alter other parts of computation. Low-rank adaptation and full-parameter RL also provide different degrees of freedom for responding to these changes.

AIS evaluates diffusion-based LLaDA alongside autoregressive Qwen models [7]. Its autoregressive experiments connect directly to the token-level formulation used here. Interpreting the denoising results requires likelihoods defined for that generation process, so those experiments broaden the empirical scope without validating the autoregressive factorization for diffusion.

Evaluation precision answers a separate question from training precision. A common BF16 evaluator, as used in KV4-RL, compares the parameters learned from different rollout distributions while holding the evaluation path fixed [17]. Evaluation through the intended low-precision sampler also includes the numerical realization of those parameters, as in TRACE's FP4 policy results [13]. Both are informative, but they should not be pooled as the same measure of policy quality.

Architecture and collection protocol further condition conclusions. MoE execution introduces discrete route choices; cache-heavy long responses expose different runtime bottlenecks from shorter generations; partial collection permits a response to contain tokens sampled under different parameter versions. A method evaluated in one of these regimes may motivate a hypothesis in another, but that transfer is not itself an experimental result.

Miles adds production-scale agentic system evidence, whereas QATFactory isolates a smaller NVFP4 QARL workload [25, 26]. Native low-precision arithmetic, simulated quantization, and BF16 execution on quantized operands are distinct regimes. In particular, HiFloat4 reports simulated FP4 RL rather than measured native-FP4 training [24].

Table 7 gives four contrasting execution configurations. The broader coverage matrix appears in Appendix B (Table 12). A configuration row describes the setting tested by a source, rather than every implementation to which its mechanism might transfer.

Table 7: Representative execution configurations, with numerical scope retained. QAT describes an adaptation procedure; its arithmetic must be stated separately.

Study	Learner path	Rollout and collection setting
TRACE [13]	Guided fake-quantized forward; BF16 backward.	NVFP4 W4A4 on routed experts and KV4 in softmax attention; other modules BF16. Asynchronous collection.
TRIAGE [12]	Native NVFP4 W4A4 forward; not a claim that all training states are FP4.	Native W4A4 forward; BF16 KV in the documented characterization. Synchronous math RL; no reported KV4 experiment.
Score Centering [10]	BF16 learner; distributional reconstruction supplies the correction.	Quantized-sampler stress tests include FP4 KV and INT8 sampler with INT4 KV. Policy-lag tests form a separate setting.
KV4-RL [17]	BF16 learner.	INT4 KV with bounded INT8 protection. Quota-based partial collection; generation and learning alternate. This is a cache-focused experiment.

Table 9 brings together four comparison cases that illuminate these interactions. Appendix A covers all twenty-nine focal works, and the companion evidence collection provides source locations for observations from all nineteen core studies.

6.2. Findings on Stability and Learning Quality

The most informative comparisons connect a changed component to both its intermediate effect and the resulting learning trajectory. Figure 11 identifies the baselines directly compared in TRACE and TRIAGE; Table 9 retains the corresponding result conditions. These cases illustrate how to read the mechanisms, while the companion evidence collection covers all nineteen core studies.

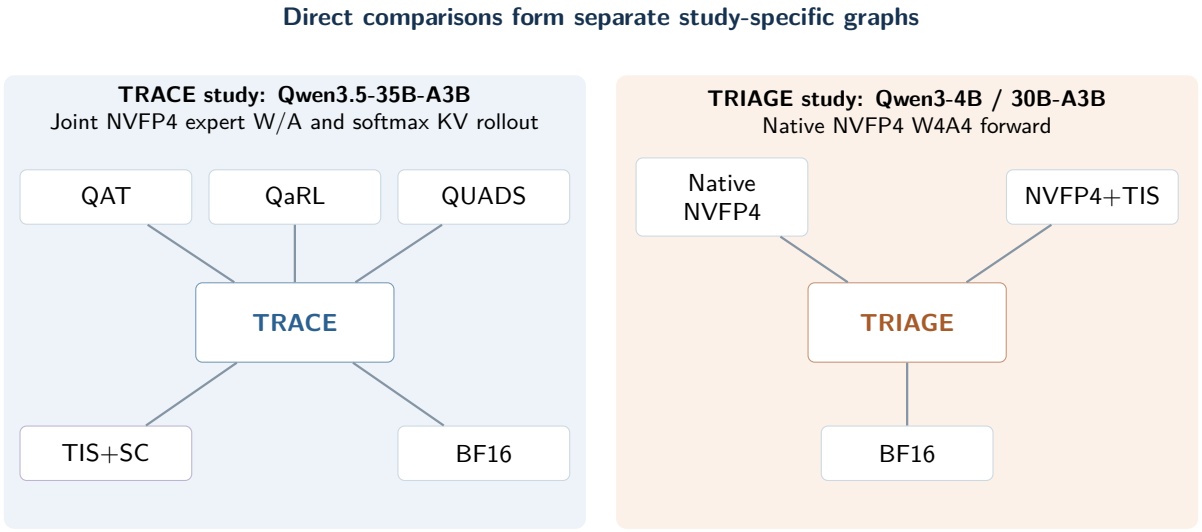


Figure 11: Direct-comparison graphs for TRACE and TRIAGE [12, 13]. Separate panels preserve the execution setting of each study; the BF16 nodes are study-specific references. Edges indicate available comparisons, not performance order. Table 9 provides the outcome and checkpoint conditions.

Reading learning trajectories. The common training diagnostics measure different objects. Policy entropy describes the spread of the next-token distribution; training reward describes the scores assigned to sampled answers; a mean absolute log-probability gap describes agreement on sampled tokens. Table 8 connects these quantities to the conclusions they support. Section 3.4 specifies the policy comparisons and reductions behind these diagnostics; here the question is how to read their joint evolution.

Table 8: Interpreting training diagnostics. Formulas illustrate common definitions; they do not reconstruct undocumented logging code. Valid-token and response reductions must be stated.

Quantity	What it measures	What it does not establish
Policy entropy H	$-\sum_a p(a) \log p(a)$ for a specified next-token policy, subsequently aggregated over prefixes.	A lower value need not indicate better learning. A label such as *entropy loss* does not establish that entropy regularization is active.
Mean reward $\bar{R}$	$B^{-1} \sum_i R_i$ over B sampled responses. With strictly binary correctness reward, it is the sampled correctness rate.	It is not the group-normalized advantage, a held-out benchmark score, or an invariant scale across reward functions.
Mean absolute gap G	$N^{-1} \sum_t \lvert d_{\theta,t} \rvert$ over N valid sampled tokens: the magnitude of learner–behavior log-probability disagreement.	It removes sign and local concentration; it is neither KL divergence nor proof of estimator fidelity or policy quality.

TRIAGE's 30B experiment illustrates the distinction: TIS keeps the aggregate gap from diverging over the measured horizon, yet its later reward and downstream scores remain below the full method [12]. Conversely, useful reward-driven updates can increase some signed gaps. Stability must therefore be read jointly with learning progress and the retained signed contributions, rather than inferred from a single discrepancy curve.

Agreement, update fidelity, and quality are distinct outcomes. The same metric can play different roles across methods. In AIS, a gap participates in a control signal; in KV4-RL, conditional KL evaluates a numerical design; in TRIAGE, signed and localized statistics motivate a directional intervention [7, 12, 17]. A method that uses a diagnostic is not automatically validated by lowering that diagnostic. Stronger interpretation asks whether its component changes the proposed intermediate effect while preserving useful learning. This also explains why a paired probe and an online curve can move differently: the former holds the policy and histories fixed, while the latter changes with the learned parameters and sampled population.

Correction and mismatch co-evolve during learning. QuRL observes disagreement beyond an initially stable period, with ACR and UAQ addressing different changes in clipping and update visibility [5]. AIS adapts the correction mixture, ACRL regulates discrepancy through update influence, and Score Centering removes a conditional estimator drift term [7, 9, 10]. Each intervention changes the policy and hence the mismatch encountered later. P3O's Qwen3-8B FP8 comparison similarly links batch-adaptive control to preserved later performance within the evaluated horizon [21]. The recurring observation is dynamic: a correction changes the learner and thereby changes the mismatch that later correction must handle.

Shared computation matters more than nominal symmetry. Jet-RL couples precision flow; QaRL, QUADS, and TRACE couple representations or realized rounding decisions [3, 6, 8, 13]. QUADS separates two effects: asymmetric weight QAT recovers much of the performance loss, and rollout activation compensation further narrows the gap. Its asymmetric-versus-symmetric comparison shows why nominal symmetry alone does not define effective alignment. HiFloat4's simulated precision ablations also locate a residual error in rollout activations [24]. FP8-RL follows a changing policy through calibration and synchronization, while QATFactory compares learning in the target representation with quantizing only the learned endpoint [4, 26]. Together, these comparisons support examining what is actually shared—operators, weights, scales, or rounding outcomes—before interpreting a claim of aligned precision.

Preserving useful updates matters beyond reducing average gaps. The full-pipeline FP8 study connects excessive lower-side clipping of negative-advantage tokens to pathological training dynamics, whereas TRIAGE uses local and direction-dependent information to select interventions [11, 12]. DRPO preserves corrective gradients outside its trust-region boundary; MIPU instead tests a whole candidate after conversion to the inference policy [22, 23]. Together these methods show why the location, direction, and stage of control matter. Aggregate clipping fractions or mean log-probability gaps can conceal these distinctions. TRIAGE's MoE routing-conditioned analysis further separates a population trend from reliable directional prediction for each token. Checkpoint selection adds another boundary: the full-pipeline FP8 study reports its main GRPO endpoints from the best average checkpoint over 500 steps [11], whereas TRIAGE evaluates stable arms at their final scheduled step. Similar endpoint scores can therefore summarize different trajectory-selection rules.

Table 9: Within-study evidence and its comparison boundary. Values belong to the stated setting; they are not pooled across studies.

Study	Controlled comparison	Reported result	Interpretation and limit
TRACE [13] 35B MoE	Joint NVFP4 W/A/KV rollout; TIS+SC uses cap 2 and top-128 reconstruction.	Four-task mean: TRACE 75.3, TIS+SC 74.1, BF16 74.9.	SC selects its best mean checkpoint through 400 steps. TRACE does not win on every task; this table reports no uncertainty interval.
TRIAGE [12] 30B MoE	BF16, native NVFP4, TIS, and full gating/repair recipe. Stable arms use step 1700.	Five-task mean: TRIAGE 70.96, TIS 63.18, BF16 72.41.	Supports recovery in this native-W4A4 setup; no SC/TIS+SC or KV4 comparison is reported.
Score Centering [10] 30B MoE	Correction rules share the objective; quantization and lag are tested separately.	With FP4 KV, SC and TIS remain stable; stronger INT8/INT4 stress separates them further.	Quantization curves report training accuracy. Shared-objective controls are not full reproductions of every baseline recipe.
KV4-RL [17] 8B dense	Cache precision varies with TIS retained; all learned policies are evaluated in BF16.	At 400 updates, six-task mean: 52.8 versus BF16's 52.4; cumulative full-iteration speedup 1.27×.	Evaluation-seed variation is not independent training-run variation; this is not a joint W4A4/KV4 comparison.

The direct comparisons answer different practical questions. TRACE tests whether rollout-guided alignment improves over existing QAT and TIS+SC in its joint low-precision configuration. Its 1.2-point mean advantage over TIS+SC is accompanied by mixed task-wise outcomes. TRIAGE tests whether direction-aware control recovers learning left unresolved by TIS in native W4A4 execution. The latter experiment does not compare score centering, so it leaves that choice open [12, 13].

Complementarity is most persuasive when a component ablation identifies the remaining error. QUADS separates weight alignment from activation compensation, QuRL addresses update visibility alongside clipping control, and KV4-RL changes cache fidelity while retaining TIS [5, 8, 17]. These examples support the combinations actually tested and motivate a broader question: after one intervention acts, which residual mechanism still limits progress?

6.3. Efficiency Gains and Implementation Costs

An intervention's computational benefit depends on the work required for generation, learning, synchronization, and correction. For serialized phases, iteration time can be decomposed as

$$T_{\text{iter}} = T_{\text{rollout}} + T_{\text{learner}} + T_{\text{sync}} + T_{\text{quant/calib}} + T_{\text{other}}, \tag{12}$$

where synchronization and quantization/calibration costs are separated from generation and learning. Accelerating one phase exposes the others. With overlapping execution, the relevant quantity is the critical path, so the phase times in Equation (12) cannot simply be added.

The serialized model also gives a simple bound on what a rollout improvement can achieve. If rollout occupies fraction f of the baseline time, a rollout speedup s and added work hT_0 give $T/T_0 = (1-f) + f/s + h$, holding the other phases and workload fixed. Figure 12 illustrates the arithmetic. Overlap, changed response lengths, or changed concurrency require a different cost model.

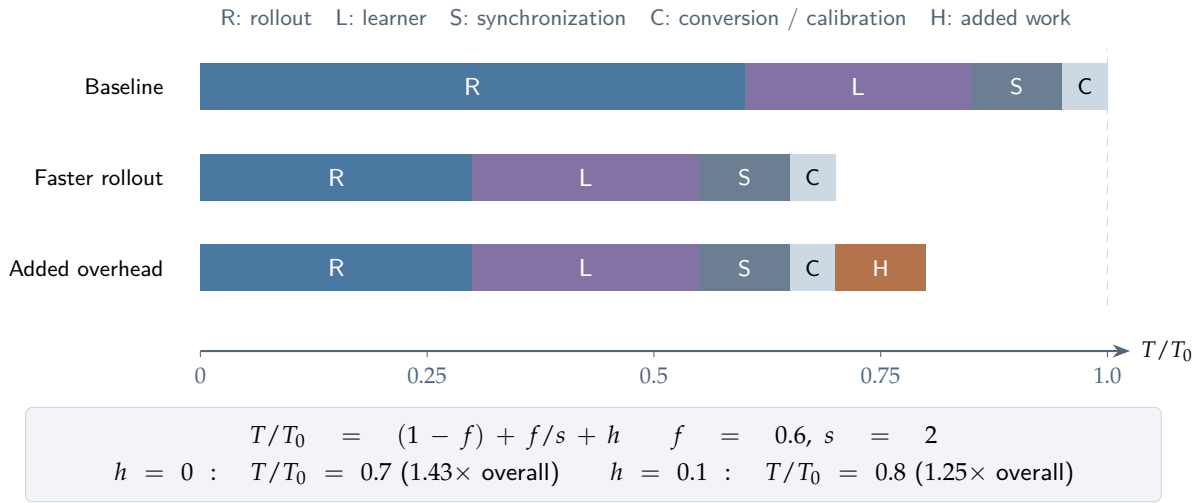


Figure 12: Serialized timing with a faster rollout phase and added overhead. Doubling rollout speed reduces iteration time from T_0 to $0.7T_0$ when rollout accounts for 60% of the baseline. Added work of $0.1T_0$ reduces the overall speedup from 1.43× to 1.25×. This is a constructed illustration, not an experimental result.

Each intervention changes a different part of this cost. FP8-RL illustrates the work needed for scale estimation and numerical conversion [4]. Reference calibration adds learner computation;

guided rounding adds storage, transport, and state matching; selective precision adds work in the protected region. Table 10 relates these dependencies to the information each method needs.

P3O requires batch-level ratio statistics, while MIPU's acceptance step adds inference-side assessment and possible rollback [21, 23]. Miles and QATFactory emphasize conversion and inference-ready weight transfer as part of the execution contract [25, 26]. TBIK's coordinated reductions can carry a kernel-performance cost [28]. Rollout-ResQ's sparse correction is designed for efficient hardware execution, but its simulated experiments do not measure a native FP4 speedup [24].

The useful quantization target depends on the workload as well as the format. QUADS targets native W4A4 expert computation, whereas KV4-RL studies how sustained concurrency and long contexts increase the value of reducing KV processing [8, 17]. Their settings motivate different numerical targets; a bit-width comparison alone does not identify the dominant cost.

Token throughput also depends on the realized workload. A policy that produces shorter responses can appear faster without improving execution per token; one that produces longer responses can incur more work despite a faster kernel. Memory savings can increase concurrency, which changes the workload seen by attention and compute kernels. Serving research such as PagedAttention provides background for why cache management and scheduling affect observed throughput [32], but serving speed alone does not establish RL learning efficiency.

Table 10: Information and execution dependencies of the reviewed approaches. These interfaces explain system cost without assigning an overhead ranking.

Approach	Information or execution dependency	Consequence for implementation
TIS / AIS / CIS	Behavior probabilities and learner/proximal probabilities	Stored probabilities must include the sampling transformations.
Score Centering	A behavior-weighted score expectation, exact or approximated	Exact evaluation requires distributional information; reconstruction introduces approximation.
Aligned QAT / shared precision	Matching quantization and conversion rules across endpoints	Precision exceptions, kernels, and weight export form part of the configuration.
Guided rounding	Realized rollout-side quantization information	Guidance requires storage, transport, and matching to learner states.
Calibrated Clipping [11]	Higher-precision reference ratios at calibration points	Reference passes add training computation, amortized across calibration intervals.
Batch-adaptive control [21]	Policy-ratio statistics aggregated over the batch	Reduction rules and the sampled batch determine the reliability estimate.
Inference-policy acceptance [23]	Assessment of the synchronized candidate	Candidate evaluation and possible rollback add work beyond learner updates.
Selective KV precision	Low-bit cache execution with protected positions	Protected positions add work whose share depends on the context workload.

KV4-RL illustrates the difference between generation and training gains. Over 400 iterations on Qwen3-8B-Base, it reports 1.80× cumulative rollout speedup and 1.27× cumulative full-iteration speedup over BF16 rollout [17]. Other phases absorb part of the generation gain. Its common-BF16 evaluation separately compares learned-policy quality under a shared evaluation precision. Both conclusions apply to the study's collection workload.

Time or compute to a specified quality level connects these measurements to the learning objective. Endpoint scores, iteration time, and rollout throughput supply different parts of the picture. A remedy can improve fidelity yet require more learner passes; a faster sampler can change response lengths or the number of updates needed. The complete trajectory is therefore needed when interpreting quality per budget or time to a target. In overlapped systems, storage and guidance traffic also compete for bandwidth, so an overhead measured in isolation need not equal its effect on the critical path.

6.4. Evidence Limits and Coverage Gaps

The strongest empirical support comes from comparisons that isolate a mechanism within a specified execution setting. Mathematical reasoning recurs across the corpus; controlled evidence is thinner for interactive trajectories, broader task distributions, and changes in hardware or kernels. Native arithmetic, quantization simulation, and BF16 computation on quantized values remain different evidence regimes.

Reporting varies at both the measurement and experiment levels. A numerical gap needs the specification in Section 3.4; a learning comparison also needs its budget, checkpoint rule, and source of uncertainty. Training seeds, negative results, and cross-hardware replication have not been systematically extracted for the full corpus. Repeated decoding at fixed parameters measures evaluation variation, while independently initialized RL runs expose training variation. Smoothing reveals a trend but supplies neither kind of uncertainty by itself. These distinctions limit the resolution of any cross-study ordering.

Further gaps concern combinations. The presence of separate W4A4 and KV4 studies does not establish their joint behavior, and asynchronous system demonstrations do not isolate every interaction between quantization and policy lag. Scheduling and diffusion results broaden the context, but require direct low-bit validation or generation-specific likelihoods before transferring the autoregressive analysis.

7. Open Problems

The synthesis leaves four connected questions: how to recognize harmful mismatch, compose remedies, identify stateful behavior, and establish transfer across settings.

7.1. Predicting Harmful Mismatch

The central diagnostic problem is to predict lost learning before reward reveals the damage. Ratio tails, conditional score drift, signed clipping errors, and segment concentration identify different candidate mechanisms. A useful predictor must distinguish harmful disagreement from legitimate policy movement and from a controller that simply discards difficult data.

The next step is to relate a diagnostic at a specified stage to subsequent retained learning under several intervention rules. Paired numerical probes can localize an execution source; fixed-batch update audits can test a directional explanation; held-out trajectories can establish whether it matters. A robust predictor would connect these levels and remain informative after the remedy itself changes the data distribution. Lower diagnostic values alone do not supply that connection.

7.2. Composing Numerical and Learning Interventions

A component's value depends on the residual error after other components act. Weight alignment can expose the value of activation compensation; estimator correction can change whether a restrictive update rule still helps. The mechanisms may be complementary, redundant, or hidden by another control.

The combinations in Table 6 provide starting points, while rollout-guided alignment with score centering and cache protection with direction-aware control remain questions for direct study. Component ablations should identify which residual effect changes, whether useful signed contributions survive, and what additional computation or state is required. This separates mechanistic synergy from a bundle whose result depends on retuning or one dominant component.

7.3. Precision, Staleness, and Long-Horizon Interaction

A checkpoint need not identify the behavior policy. Responses can cross synchronization boundaries, and retained KV entries encode earlier computations. Applying the separation in Equation (2) therefore requires the relevant history. What is the smallest sufficient behavior record: token probabilities, version identifiers, cache provenance, or additional execution state?

Early numerical perturbations also change the prefixes visited later, beyond the conditional correction at an observed token. Precision allocation across layers, positions, and training phases should therefore follow learning benefit rather than tensor error alone. Multi-turn environments extend this question to tool observations and state transitions.

7.4. Generalization and Comparable Evidence

Portability depends on the information a mechanism needs. Action reweighting requires actual decoding probabilities; score reconstruction requires broader distributional information; rounding guidance requires paired numerical states. Engine, generation, or synchronization changes can invalidate these dependencies without changing the format label.

Component studies should isolate compatible corrections under a shared execution path and base objective. Complete-system comparisons should preserve each method's essential design and assess quality and total cost under matched resources. Fixed-budget quality and cost to a predeclared target connect progress to runtime; failure to reach the target remains an outcome. These principles motivate a future benchmark outside the present survey's empirical scope.

8. Conclusion

Low-precision rollout changes the experience supplied to RL and the policy updates computed from it. The four intervention classes locate remedies in computation alignment, numerical fidelity and update visibility, gradient estimation, and update control, complemented by scheduling.

The central distinctions are between numerical accuracy and cross-path agreement, estimation and update control, and faster execution and more efficient learning. Measurement practice connects them: explicit policies, populations, and stages make a discrepancy interpretable; evidence on retained learning and total cost establishes its value. These connections guide combinations that improve a numerical policy while preserving the experience and updates it needs to learn.

A. Surveyed Papers and Section Mapping

Table 11 summarizes the twenty-nine focal studies, numerical regimes, and component assignments. Years denote first public appearance; bibliography records may cite later formal publications. Main discussion locations precede additional locations in parentheses. Contribution roles are defined in Appendix B.

Table 11: Surveyed papers: numerical regimes, method components, and section mapping. Conference and preprint versions count as one contribution.

Work / first year	Numerical regime	Component(s) and contribution	Discussion
Core method and system studies (19)			
FlashRL / TIS [1] 2025	Higher-precision learner; quantized rollout	**B1**: Recorded behavior probabilities and truncated correction.	**5.1** (1.1; 2.1)
Jet-RL [3] 2026	Unified FP8 learner and rollout paths	**A1; A2**: Unified FP8 precision flow.	**4.1** (6.2)
FP8-RL [4] 2026	W8A8 rollout; FP8 attention / KV; FP8 learner explored in v2	**A2; system**: W8A8 rollout, FP8 attention/KV, and stack integration.	**4.2** (5.1; 6.2; 6.3)
QuRL [5] 2026	INT8 / FP8 rollout	**B2; A2**: Adaptive clipping and update-aware quantization.	**5.2** (3.1; 4.2; 5.4; 6.2)
QaRL [6] 2026	Quantized MoE rollout; aligned learner QAT	**A1; B2**: Rollout-aligned QAT and trust-band updates.	**4.1** (5.2; 5.4; 6.2)
P3O / ESS [21] 2026	BF16 learner / FP8 rollout; dense reasoning models	**B2**: Batch-adaptive ESS cap and regularization.	**5.2** (6.2; 6.3)
AIS [7] 2026	BF16 learner / FP8 rollout; autoregressive and diffusion models	**B1**: Adaptive importance-weighted estimator mixture.	**5.1** (6.1; 6.2)
DRPO [22] 2026	BF16, FP8 rollout-only, and FP8 end-to-end MoE settings	**B2**: Smooth corrective divergence regularization.	**5.2** (6.2)
MIPU [23] 2026	BF16 learner / FP8 rollout; Qwen3-1.7B and 4B	**B2; formulation**: Inference-policy candidate construction and acceptance.	**5.2** (2.1; 6.2; 6.3)
QUADS [8] 2026	NVFP4 MoE rollout; learner weight QAT	**A1; A2**: NVFP4 MoE weight alignment and activation compensation.	**4.1** (4.2; 5.4; 6.2)
ACRL [9] 2026	FP8 rollout; discrepancy-dependent updates	**B2**: Discrepancy-dependent update control.	**5.2** (3.3; 6.2)
HiFloat4 / Rollout-ResQ [24] 2026	Simulated end-to-end HiF4 / MXFP4; rollout residual correction	**A2**: Sparse residual correction for FP4 rollout activations.	**4.2** (3.1; 6.2; 6.3)
Miles v0.1 [25] 2026	FP8 / MXFP8 / NVFP4 RL recipes; precision exceptions	**A1; A2; system**: Shared quantization, synchronization, and precision contracts.	**4.1** (4.2; 6.1; 6.3)
Score Centering [10] 2026	BF16 learner; FP4/INT4 KV stress and separate lag tests	**B1**: Additive behavior-weighted score correction.	**5.1** (2.2; 3.2; 5.4; 6.2)
Full Pipeline FP8 RL [11] 2026	FP8 learner and rollout; calibration against BF16	**B2; system**: FP8 learner/rollout and calibrated clipping.	**5.2** (3.2; 6.2; 6.3)

Continued on next page

Table 11 continued: Surveyed papers

Work / first year	Numerical regime	Component(s) and contribution	Discussion
QATFactory / QARL [26] 2026	Native NVFP4 rollout; simulated-quantization BF16 training	**A1; system**: Deployment-aligned QARL and inference-ready weight synchronization.	**4.1** (6.2; 6.3)
TRIAGE [12] 2026	Native W4A4 forward; documented BF16 KV; synchronous RL	**B2**: Direction-aware control for native NVFP4.	**5.2** (3.3; 6.2)
TRACE [13] 2026	NVFP4 expert W4A4 + softmax KV4; guided QAT, BF16 backward	**A1**: Rollout-guided FP4 rounding for MoE.	**4.1** (3.1; 6.2)
KV4-RL [17] 2026	BF16 learner; INT4 KV with bounded INT8 protection	**A2; system**: INT4 KV with selective INT8 protection.	**4.2** (5.1; 6.2; 6.3)
Supporting formulation, theory, and general TIM studies (9)			
R3 [15] 2025	MoE routing; general execution mismatch	**Support; A1**: MoE routing alignment.	**3.1** (4.1)
FP16-Mismatch [16] 2025	BF16 / FP16 numerical comparison	**Support; A2**: BF16/FP16 precision-selection evidence.	**3.1**
TBIK / Deterministic Inference [28] 2025	BF16 / FP32; matched reduction order across TP sizes	**Support; A1**: Cross-TP deterministic kernels as an alignment reference.	**3.1** (4.1; 6.3)
MiniRL [14] 2025	Hybrid-engine RL; includes FP8-rollout experiments	**Support; B1/B2**: Policy roles and mismatch-aware formulation; FP8 experiments.	**2.1** (1.2; 5.1)
Beyond Precision [18] 2026	General TIM; optimization scheduling	**Support; B3**: Optimization schedule under general TIM.	**5.3** (3.3)
DPPO [29] 2026	General LLM trust-region and cross-engine mismatch evidence	**Support; B2**: Divergence-based trust regions under general TIM.	**5.2** (3.2)
VeXact / Diagnosing TIM [19] 2026	Matched-execution diagnostic reference	**Diagnostic**: Zero-mismatch diagnostic reference.	**3.4**
CIS [20] 2026	BF16 cross-engine mismatch	**Support; B1**: Confidence-dependent correction under BF16 cross-engine mismatch.	**5.1** (3.2)
Baselines and Score Centering [27] 2026	Sequence- and token-level regularized importance sampling	**Support; B1**: Granularity-dependent roles of baselines and score centering.	**5.1** (5.4)
Exploration boundary case (1)			
QeRL [2] 2025	NVFP4 with LoRA and adaptive noise	**Boundary**: NVFP4 with LoRA and adaptive noise; exploration setting.	**Box 1**

B. Source Selection and Version Handling

Construction of this snapshot. The collection comprises twenty-nine focal contributions on quantized rollout, low-precision RL, and the mechanisms needed to interpret training–rollout mismatch. Primary papers and official records provide the evidence. A targeted update searched arXiv metadata through October 20, 2026, followed by primary-text checks and backward citation tracing. This curated narrative review does not claim exhaustive coverage.

Inclusion and exclusion. A focal study contributes a direct quantized-rollout or low-precision RL intervention, a mechanism needed to interpret that setting, or a carefully identified boundary case. General quantization and RL background is cited selectively. Papers devoted only to low-precision pretraining, unrelated generation modalities, or alternative rollout acceleration do not become focal entries solely because they use the same precision format.

Versions and names. The catalog distinguishes first-publication dates, publication venues, and reviewed versions. QeRL, QuRL, and QaRL use formal publication records in the bibliography. Conference and preprint versions, aliases, and changes in presentation count as one contribution. Unpinned arXiv links are distinguished from locally reviewed versions, since a later revision may change the claims supported by a source.

Attribution of evidence. Method claims and observations are attributed to primary sources. The likelihood, drift, and clipping identities explain mechanisms under the stated assumptions; they are distinct from experimental findings. Figures and tables are original syntheses, and constructed examples are labeled separately from published measurements. The companion evidence collection links observations to reviewed sources.

Evidence roles and verification. Contribution role identifies relevance to the argument; evidence domain identifies the setting examined. Neither grades experimental quality. Source-linked observations retain their conditions and limitations. No independent reproduction was performed, and unextracted statistical or implementation fields remain unassessed. The catalog separates identified reviewed versions from version-unspecified records.

The coverage matrix below records the regimes represented in the collection. Its dimensions describe scope; they do not define method classes or experimental quality.

Table 12: Coverage matrix for the qualitative snapshot. Rows are independent dimensions; the presence of two regimes does not imply that their combination has been evaluated.

Dimension	Represented in the focal collection	Boundary of the synthesis
Intervention	A1/A2 numerical design; B1/B2 learning rules; complementary scheduling.	Direct low-bit evidence is thinner for B3 than for the other classes.
Numerical interface	Weight/activation quantization, KV storage, scales, rounding, routing, and synchronization.	An interface-level result does not establish full-network agreement.
Precision	FP8, FP4/NVFP4, INT8/INT4; BF16/FP16 comparisons provide context.	Format labels omit scale granularity, precision exceptions, and native versus simulated arithmetic.
Model and adaptation	Dense and MoE models; low-rank and full-parameter updates appear in different studies.	Architecture and adaptation are separate axes; cross-regime transfer is not assumed.
Collection	Snapshot-based hybrid execution, policy lag, and quota-based partial collection.	Stateful provenance and fully overlapped execution require additional behavior and timing semantics.
Learning and generation	PPO/GRPO-style autoregressive RL; AIS also studies diffusion generation.	This survey's likelihood equations are autoregressive; denoising requires its own formulation.
Contribution role	Nineteen core studies, nine supporting studies, and one exploration boundary.	Role identifies relevance to the argument, independently of experimental strength.

C. Evidence, Assumptions, and Transfer

Transfer depends on the relationship between sampled behavior and learner computation. Parameter versions, numerical paths, decoding, and retained state determine the behavior law; learner, proximal, and reference policies define separate loss roles. Changing execution can alter a correction's meaning even at the same nominal precision.

Each approximation leaves a specific residual: prefix selection after conditional correction, truncation bias, unmatched execution after alignment, or reconstruction error in an expectation. A tested combination supports its reported estimator and regime; transfer requires the assumptions responsible for that result to remain valid.

Conditional and complete-response correction. The token-level identity in Equation (3) holds at a specified prefix. Recovering a response-level expectation additionally requires correcting how often that prefix is reached.

For comparison, the full-response likelihood ratio is

$$W_\theta(y \mid x) = \frac{P_\theta(y \mid x)}{Q(y \mid x)} = \prod_{t=1}^{T} \frac{p_\theta(y_t \mid h_t)}{q_{k(t)}(y_t \mid h_t)}. \tag{13}$$

Here P_θ and Q are the induced response laws, and the denominator factors must be their actual conditional probabilities under the documented protocol. Unobserved execution randomness requires marginalizing that state or extending the likelihood to the corresponding execution events. The two response laws use a common stopping convention; an end-of-sequence decision is included when applicable.

For a fixed behavior law and a parameter-independent reward R, the usual likelihood-ratio identity gives

$$\nabla_\theta \mathbb{E}_{P_\theta}[R] = \mathbb{E}_Q\left[W_\theta R \sum_{t=1}^{T} s_\theta(y_t, h_t)\right] \tag{14}$$

under support, integrability, and differentiation assumptions. The product illustrates why exact response-level correction can have high variance; multiplying only the current token ratio into an arbitrary sampled advantage is a different estimator. Nor does Equation (14) directly justify reusing group-normalized advantages computed under Q. Score Centering explicitly distinguishes sequence-level importance sampling from practical token-level corrections [10].

A two-step example in Figure 13 makes the distinction tangible. The second-token conditionals agree exactly, yet the first-token histories have different probabilities. A correction using only the second-token ratio therefore leaves that prefix imbalance, whereas the full-response ratio includes the earlier decision.

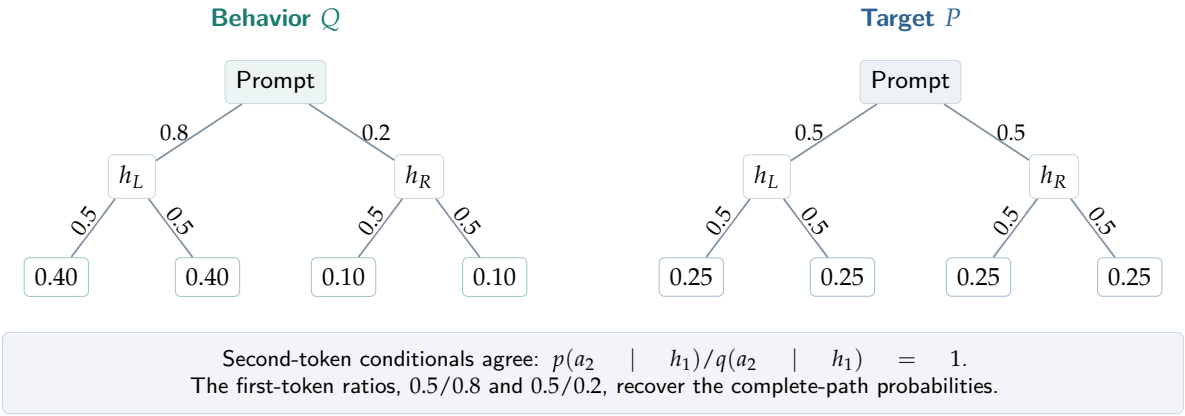


Figure 13: Conditional and response-level correction in a two-step probability tree. Leaf values are complete-response probabilities. Equal second-token conditionals coexist with different prefix masses; the product ratio restores the four target probabilities of 0.25. This is a constructed illustration, not an experimental result.

D. Terminology

Table 13: Terminology and component abbreviations used in the text and figures.

Term	Meaning
TIM	Training–inference mismatch; the survey emphasizes the learner–rollout interface.
W / A / KV	Weights, activations, and key–value cache; distinct numerical targets.
MoE / LoRA	Mixture-of-experts architecture / low-rank adaptation; architecture and adaptation are separate dimensions.
FP8 / FP4 / NVFP4	Floating-point format families with eight or four bits; NVFP4 is a particular four-bit representation. Scaling, accumulation, and exceptions also determine execution.
INT8 / INT4	Integer quantization at eight or four bits, with scales and execution semantics specified separately.
QAT	Quantization-aware training: the learner forward path models a quantized representation; this need not reproduce native kernel arithmetic.
TIS / MIS	Truncated / masked importance sampling; bounded or discarded corrections alter the estimator.
AIS / CIS	Adaptive / calibrated importance sampling: estimator-mixture adaptation and confidence-dependent truncation, respectively.
ACR / UAQ	QuRL's Adaptive Clipping Range / Update-Aware Quantization; B2 update control and A2 update visibility.
TBPO / RAC	QaRL's Trust-Band Policy Optimization / QUADS's Residual Activation Compensation; distinct objective and numerical components.
PPO / GRPO	Proximal Policy Optimization / Group Relative Policy Optimization. Their loss roles require behavior and proximal policies to be distinguished.
ESS / P3O	Effective sample size / Policy-on Policy-off Policy Optimization; batch-ratio reliability controls update influence and regularization.
DPPO / DRPO	Divergence Proximal / Divergence Regularized Policy Optimization; divergence-based masking versus smooth corrective regularization.
MIPI / MIPU	Monotonic Inference Policy Improvement / Update; inference-policy objective and practical candidate construction/acceptance.
Rollout-ResQ	Sparse residual correction of rollout activation quantization; the reviewed HiFloat4 experiments simulate FP4.
TBIK	Tree-Based Invariant Kernels; matched reduction order across tensor-parallel sizes.
Partial rollout	Collection in which a response can span collection boundaries; it does not by itself imply concurrent generation and optimization.
Cache provenance	Parameter versions and numerical execution paths that produced retained cache entries.
Behavior / proximal / reference	Actual sampling law / update-ratio anchor / KL-regularization target; see Table 1.

E. Reported Mismatch Measurements

Table 14 summarizes selected diagnostic evidence from every core method/system study. It distinguishes explicit policy-distance definitions, control variables, local numerical probes, and learning outcomes. A field left unspecified in the cited description is not inferred from another paper's similarly named metric. The companion repository supplies full titles, versioned source links, and detailed interpretation boundaries.

Table 14: Measurements and comparison boundaries across the nineteen core studies. Locations refer to each study's reviewed version. This is a map of selected evidence, not an exhaustive inventory of logging code.

Study	Reported observable and location	Comparison boundary
FlashRL / TIS [1]	Cross-engine KL (Fig. 3); TIS uses sampled-action ratios.	The plotted KL direction and reduction are not specified in the caption.
Jet-RL [3]	Length- and task-conditioned learning curves (§ 3.2).	Outcome evidence; these curves do not isolate a numerical policy distance.
FP8-RL [4]	Rollout-to-training mismatch KL (§ 2.2.1; Fig. 12).	Direction is explicit; estimator and reduction require further implementation detail.
QuRL [5]	Behavior-to-proximal KL; maximum proximal/behavior ratio (Fig. 3).	Average discrepancy and rare large weights are distinct; neither is the reference KL penalty.
QaRL [6]	Proximal-ratio and mismatch-weight distributions (Fig. 4); response trust bands.	Keep the two ratios separate; geometric-mean response ratios differ from likelihood products.
P3O / ESS [21]	Valid-token normalized ESS, using current/behavior ratios (Eq. 11).	Measures concentration; includes policy change and does not identify numerical error alone.
AIS [7]	Mean absolute log gap; capped-weight ESS; advantage-weight variance amplification (§ 3.1).	These govern distinct gates; capped and raw ESS are different observables.
DRPO [22]	Sampled-action Binary-TV geometry; learning and entropy trajectories (§ 3; App. D.4).	The constraint is not full-vocabulary TV or, by itself, an isolated TIM measurement.
MIPU [23]	Sampled-token mismatch K3; inference-gap proxy and rollback rate (§ 5).	KL estimation and candidate acceptance assess different stages of an update.
QUADS [8]	Maximum log gap; layer-wise MoE L2 discrepancy (Figs. 6–7).	Local tensor error and policy-level extremes have different scales and comparison populations.
ACRL [9]	Absolute probability gap, response/group averaged relative to baseline X (Eqs. 8–12).	Distinct from log gaps; the baseline-relative controller does not define a universal tolerance.
HiFloat4 / Rollout-ResQ [24]	Residual sparsification MSE/RMSE and precision ablations (Fig. 4; § 5).	Tensor reconstruction error should not be relabeled a learner–sampler policy distance.
Miles v0.1 [25]	Same-token log-probability checks; reference-run KL; exact-mode zero difference (§ 3.1; § 5.3; Fig. 5).	Different modes use different comparisons; exact sampled-token scores do not establish full-distribution equality.

Continued on next page

Table 14 continued

Study	Reported observable and location	Comparison boundary
Score Centering [10]	Conditional score-mean analysis; controlled learning curves under noise, quantization, and lag (§§ 3–5).	Distinguish the analytical drift term from logged measurements and deliberately amplified stress.
Full Pipeline FP8 RL [11]	Probability/ratio NMAE against BF16 shadows; sign-separated over-clipping (§ 3).	Counterfactual branch errors differ from ordinary clipping rates and fixed-snapshot TIM.
QATFactory / QARL [26]	QARL learning curves and exported-policy quality (§ 4.6).	Held-out teacher KL elsewhere in the paper belongs to QAD, not an online RL TIM measurement.
TRIAGE [12]	Mean absolute gap, signed tails, directional masses, and localization (§ 3; App. B).	30B characterization is routing-conditioned; its raw magnitude is not directly comparable with ordinary 4B gaps.
TRACE [13]	Log-gap maxima/minima; relative excess train–rollout activation MSE (Figs. 4, 11).	Extremes differ from means; excess MSE uses BF16 cross-path MSE as its denominator.
KV4-RL [17]	Shared-prefix full-distribution BF16-to-quantized KL (Eq. 5.2; App. B.5).	Same checkpoint and pre-decoding probabilities isolate conditional KV distortion; trajectory shift remains outside the metric.

From a plot label to a reproducible quantity. The following guidance is distilled from the source practices above; it is not an established common protocol or a new benchmark. A measurement specification should make the population and comparison recoverable without inferring them from the training objective.

Three quantities often called KL. A learner–sampler KL diagnoses an execution or data-distribution difference. A current–proximal KL describes policy movement during optimization. A learner–reference KL constrains departure from an external anchor. Their denominators, parameter snapshots, and sample laws differ. A fourth use, teacher–student KL in quantization-aware distillation, measures model fidelity rather than online RL mismatch; QATFactory's QAD and QARL experiments illustrate this distinction [26]. A dashboard label such as *actor KL* is therefore incomplete without its probability pair and estimator.

A minimal paired analysis. For an online diagnostic, retain the generation-time score and relevant behavior version, then state which learner replay supplies the numerator. To measure one update's motion, replay the same observed prefixes before and after that update while preserving the original behavior scores; TRIAGE uses this design in its directional audit [12]. A current-learner/old-behavior comparison remains a combined operational gap unless a matched sampling-snapshot learner is also available. Numerical probes should separately state which kernels, route decisions, and retained cache states are shared. None of these comparisons reconstructs trajectories that were never sampled.

Table 15: Practical reporting choices for mismatch diagnostics. Examples retain their source-specific scope; the final column states what must travel with the reported number.

Measurement choice	Source practice or example	Required interpretation context
Execution-only comparison	KV4-RL fixes checkpoint and BF16 prefixes for full-distribution KL [17].	Prefix source; raw versus decoded probabilities; routes and cache state; KL direction.
Update-time comparison	TRIAGE replays an identical batch before/after an update with generation log-probabilities held fixed [12].	Learner versions; pre/post stage; fixed advantages and masks; route replay.
Collection-time record	CIS logs generation scores, start-of-update learner scores and rollout versions [20].	Decode versus prefill rescoring; version lag; stored numeric precision and sample selection.
Batch reliability	P3O aggregates raw-ratio moments; AIS uses truncated weights for its ESS gate [7, 21].	Weight definition, cap, valid-token count, and cross-rank reduction.
Decision disagreement	Full-pipeline FP8 compares clipping decisions with BF16 shadow ratios [11].	Reference path; advantage sign; boundary; ordinary versus counterfactual clipping.
Uncertainty and time axis	KV4-RL uses response-cluster bootstrap intervals for its paired fidelity probe [17].	Independent unit: token, response, evaluation seed, or training run; updates, sampled tokens, or elapsed time.

A compact diagnostic set. A useful report combines a central gap statistic with signed tails or quantiles, identifies the concentration of the actual correction weights, and examines the update decisions most closely tied to the proposed mechanism. Report these on a stable reference population as well as any selected subset, and connect them to held-out quality and total cost. Additional metrics should resolve a mechanism or ambiguity rather than accumulate dashboard curves. The relevant set depends on the intervention: paired activation error for rounding alignment, raw/capped weight moments for estimation, or sign-conditioned branch disagreement for clipping calibration.

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